UNIVERSITI TEKNOLOGI MALAYSIA

PROJECT 1

SYSTEM ANALYSIS AND DESIGN

NO	NAME	MATRIC NO
1	MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI	A23CS0250
2	MUHAMMAD NAIM BIN ABDULLAH	A23CS0134
3	SYARIFAH DANIA BINTI SYED ABU BAKAR	A23CS0183
4	YASMIN BATRISYIA BINTI ZAHIRUDDIN	A23CS0201

Table of content

3.4 Project Planning Schedule (* include in phases of the methodology)

3.4. 1 Findings from feasibility study (background study)
- 3.4.2 Cost Benefit Analysis (CBA)
- 3.4.3 Work-based Structure (WBS)
- 3.4.4 Pert Chart
- 3.4.5 Gantt Chart
- 3.4.6 GitHub

3.5 Summary

Chapter 4: System Analysis and Design

4.1 Introduction

4.2 System Requirements Gathering Techniques

- 4.2.1 Method used
- 4.2.2 Summary from method used (include example Interview/Questionnaire/Observation)
- 4.2.3 Current business process (scenarios, workflow)
- 4.2.4 Company (Cooperative of KADA)
- 4.2.5 Logical DFD AS-IS system (Context Diagram, Diagram 0, Child)

4.3 System Requirements

4.3.1. Functional Requirements

4.3.2. Non Functional Requirements

4.4. System Design

4.4.1 Proposed system: Logical DFD (TO BE) System (Context Diagram, Diagram 0, Child, Process Specification)

4.4.2 Proposed system: Physical DFD (TO BE) System

4.4.3 Process specification

4.4.4 Partitioning

4.4.5 CRUD Matrix

4.4.6 Event Response Table

4.4.7 Structure Chart

4.4.8 System Architecture

(Explain each component in system architecture)

(Such as web page, web server etc)

4.4.9 Interface Design (Input/Output Design)

4.5 Summary

Chapter 1: Introduction

1.1 Overview

In this project, we are excited to collaborate with the cooperative Lembaga Kemajuan Pertanian Kemubu (KADA) to develop a computerized registration and loan application system. Currently, all of KADA's procedures are manual, where the applicants must complete the form and submit it in person. This manual method takes a lot of time and is again easy to make mistakes. The clerks need to manually enter data and handle applications, resulting in inefficiencies and delays.

We propose to digitize these processes by creating a computerized system that allows users to complete their registration and loan applications online. The new system will feature a user-friendly interface where candidates will be able to electronically fill out forms, upload documents, and enter their personal information. Authorized KADA staff will have efficient access to this securely stored data in a centralized database for review and processing. The efficient review process, automated notifications, and application status tracking features of the loan application system would greatly shorten the time needed for approvals and increase applicant satisfaction.

The implementation of this online system will bring numerous benefits to KADA and its members. It will minimize human data input errors, cut down on paperwork, and improve operational efficiency. With faster processing times and more flexibility, the new system will increase member satisfaction and perhaps get more people to apply for loans and join the cooperative. This transformation will not only simplify administrative processes but also contribute to the growth and success of KADA.

1.2 Background Study

KADA (Lembaga Kemajuan Pertanian Kemubu) is an agriculture organization in Kelantan. The workers under KADA are being given the opportunity to become cooperative members of KADA and make a loan under KADA too. But, the current system, or the as-is system in KADA to provide such services is being done manually and there is where we saw that it has very low efficiency and productivity. According to the list below, are the services that KADA provided to their staff currently.

1. Registration to become a membership of cooperative of KADA

In the current system, all the registration procedures have to undergo a manual process, such as filling in the paper form. For example, users must pick up the form in person at the headquarters in order to fill it in. In the registration process, most of the personal information of the users that are private and sensitive will be collected. Then, the users have to submit it manually also by submitting it at the headquarters. After the filled in and submitting process is done, then the registration paper would be further reviewed by the staff itself manually whether the users are qualified to become their member or not.

2. Applying a loan in cooperative of the KADA

Same as the registration process to become a membership of KADA, applying for a loan is also by filling in the paper form manually. The difference is in applying for a loan form, it would ask more detail about the user income information and guarantor. After the users have submitted it into the headquarters, it would be reviewed by the staff and the approval would be given by the higher position. The approval is given by conducting a monthly meeting between the staff and the higher position.

The current system has a lot of disadvantages, that is why we proposed a solution by making all these services online. By having it online, it could save a lot of time for the users and KADA, and it could also increase the KADA work efficiency

1.3 Problem Statement

Based on our analysis, we've identified several issues in the current system that contribute to its inefficiency and reduce its effectiveness in performing tasks smoothly and efficiently.

1. Loss of information
   Most of the records are stored in paper such as the registration documents and also the loan documents. Paper documents can be easily damaged by water, fire or mishandling. Once damaged, the information on them may become unreadable or entirely lost. Other than that, paper documents can be misplaced and lost due to human error.

2. Accumulation of data when access or update
   The clerk must locate the paper back in order to update the data, which is a difficult operation because they must scan and identify each paper in their safe in order to locate the particular one. As a result, it will take longer to finish a single task—updating the user's information.

3. Highly dependent on human
   Using paper forms greatly reduces productivity and efficiency because handling, filing, and finding these documents take a lot of time and effort. The dependence on paper slows down operations, causes delays, and keeps employees from focusing on more important tasks, ultimately making the organization less productive.

4. Difficulty in distributing and receiving the application
   In order to obtain the form and submit it, applicants must make an extensive journey to the headquarters. They are certain to be burdened by this since they will need to find time to travel to the headquarters and spend money on transportation

1.4 Proposed Solution

The system will serve as a cutting-edge substitute for the outdated paperwork system.Thus, the system will perform the same basic tasks as the original paperwork system, including handling the registration form for the new member and also the registration form to apply for a loan for the member of the organization. On the other hand, The shift from physical documentation to an online form is the main upgrade for the system. In the past, the organization primarily employed paper in their operations to store user information and to verify users during the loan application and registration processes. Initially, the current system only had three primary user views: the applicator, Ahli Lembaga Pengarah (ALK), and the clerk. However, after a few discussions, we concluded that the admin roles would be essential for the new online system.

Firstly, from the view of the applicants, The new system will definitely be beneficial for the user as the online system offers many benefits for the applicants. With the current paper system, people have to travel to the headquarters to get and submit forms, which takes a lot of time and money. This can especially be difficult for those who live far away or have busy schedules. In contrast, the online system allows people to access and submit forms anytime and from anywhere with an internet connection. This means they don't need to make a special trip, saving both time and travel costs. The online system is available 24/7, making it much more convenient. It also provides instant confirmation when an application is submitted, so applicants don't have to worry whether their forms are received by KADA or not. Additionally, online forms often help prevent mistakes by checking for missing or incorrect information before submission. Overall, the online application system makes the process faster, easier, and cheaper for everyone involved.

Secondly, for the members of the Board of Directors (ALK), their job of checking membership and loan applications becomes much easier with the new online system. Before, this process was slow and difficult because they had to be at the office and deal with lots of paperwork, which limited their flexibility and caused delayed decisions. With the online system, ALK members can do their work from anywhere and at any time, without needing to travel or handle physical documents. This means they can approve applications faster and more easily. The system also provides instant updates and notifications, making their work more efficient. As a result, the organization becomes much more productive in completing tasks that used to take days or weeks in a much shorter time. This makes the work of ALK members easier and improves the overall efficiency and speed of the organization.

The discussion revolves around creating a user-friendly system interface for clerks to handle applications smoothly. This interface allows clerks to easily access applicant details and payment records, and to update application statuses promptly. A key task for clerks is to review submitted forms, ensuring they meet the requirements set by KADA. With the system's filtering features, clerks can quickly sort through applications, accepting those that meet criteria and rejecting incomplete or non-compliant ones. By diligently updati-ng database records, clerks maintain accurate information for organizational use. This streamlined process helps clerks efficiently manage applications, contributing to the organization's overall effectiveness.

1.5 Project Objective

1. To analyze user requirements for this project.
2. To develop a detailed project plan that outlines the timeline and resources for the project.
3. To develop a comprehensive system design that outlines the requirements.

1.6 Project Scope

This project aims to upgrade KADA's current manual system to a modern online platform. The new system will make it easier to handle membership registrations and loan applications. It will improve the convenience for applicants, flexibility for ALK members, and efficiency for clerks, ultimately boosting productivity and user satisfaction.

- Enable applicants to access and submit registration and loan forms online from any location with an internet connection.
- Provide instant confirmation of form submissions to applicants, ensuring transparency and reducing errors.
- Allow ALK members to review and approve membership and loan applications remotely.
- Provide clerks with a user-friendly interface to access applicant details and payment records.

1.7 Benefit of proposed system

The project aims to transition KADA from a manual, paperwork-based system to a modern online platform. This transition will enhance the efficiency and productivity of KADA by streamlining the handling of registration forms for new members and loan applications. The planning phase involves defining the project scope, objectives, and deliverables to ensure a clear roadmap for development and implementation.

During the analysis phase, the needs of various stakeholders, including applicants, ALK members, and clerks, are thoroughly examined. The current challenges faced with the manual system are identified, such as the time-consuming nature of physical form submissions and the inefficiencies in managing paper documents. Requirements are gathered to ensure the new system addresses these pain points, providing easy access, instant submission confirmations, and remote capabilities for ALK members.

In the design phase, the focus is on creating a user-friendly interface that caters to the needs identified in the analysis phase. The system will feature an online platform where applicants can submit forms from anywhere, ALK members can review applications remotely, and clerks can efficiently manage application details and statuses. The design includes robust filtering features for clerks to sort applications and instant updates and notifications to streamline workflows. The system will also incorporate strong security measures to protect sensitive data and ensure accurate data management.

1.8 Summary

This planning, analysis, and design process ensures that the new online system will be a significant improvement over the manual processes, leading to higher productivity and greater user satisfaction across all levels of KADA.

Chapter 2: Literature Review

2.1 Introduction

Information Systems (IS) are integral components of modern organizations, facilitating the collection, storage, processing, and dissemination of information critical for decision-making processes. These systems encompass hardware, software, data, procedures, and people, working together to support organizational activities. The accessibility and usability of IS are paramount, ensuring that stakeholders can efficiently retrieve and utilize information to enhance operational efficiency and strategic planning.

Security is a fundamental aspect of IS, safeguarding sensitive data from unauthorized access, breaches, and cyber threats. Proper planning and structured analysis and design methodologies are essential in the development and implementation of effective information systems. By aligning IS with organizational goals and objectives, businesses can leverage technology to streamline processes, improve communication, and gain a competitive edge in the market.

The impact of IS on organizations is profound, influencing decision-making, resource allocation, and overall performance. Various types of information systems, such as Management Information Systems (MIS), Decision Support Systems (DSS), and Enterprise Systems (ES), cater to specific business needs and functions. Understanding the diverse capabilities of these systems is crucial for organizations seeking to optimize their operations and adapt to evolving technological landscapes.

2.2 Information system

2.2.1 Definition of information system

A system which assembles, stores, processes and delivers information relevant to an organization, in such a way that the information is accessible and useful to those who wish to use it, including managers, staff, clients and citizens. An IS is human activity (social system) which may or may not involve the use of computer systems

2.2.2 Cooperative information system

A cooperative information system is a system that facilitates collaboration and sharing of information among different entities or components to achieve common goals. These entities include individuals, organizations, software agents, or systems. A cooperative information system for KADA is an integrated digital platform designed to streamline and enhance the management of cooperative operations and member interactions. This system encompasses a wide array of functionalities including member registration, profile management, loan processing, and communication tools. By centralizing all member-related data and automating routine administrative tasks, the system ensures data accuracy, operational efficiency, and improved member services. Additionally, it provides valuable analytical insights that help tailor cooperative programs to meet member needs effectively. The implementation of such a system supports KADA Koperasi's mission to foster economic growth and community development by delivering seamless, efficient, and member-centric services.

2.3 Previous research system

Based on the OSSAD framework, Selmin Nurcan and colleagues' "A Method for Cooperative Information System Analysis and Design" offers an organized method for assessing and creating cooperative information systems in businesses. This methodology consists of multiple functionally specific descriptive models. The organizational hierarchy's roles, divisions, and resources are depicted using the Roles Descriptive Model. The Procedures Descriptive Model emphasizes organizational functionality by concentrating on the connections between procedures and resources. The Operations Descriptive Model illustrates task sequences and performers and offers comprehensive descriptions of operations. In order to automate workflow activities, the Workflow Analysis Method identifies participants, roles, communication channels, and tasks. Finally, network analysis is used by the Communication Models to explain how information moves across an organization.

The second article is “Implementation of Agile Methods on Development of Saving Loan Cooperative Information System” by Robi Setiawan and Irman Effendy, combining the Agile methodology with conventional research techniques. After gaining conceptual foundations through literature studies, data is gathered through observation and interviews. The Agile technique ensures constant input and adaptability through its iterative steps of planning, designing, developing, testing, deploying, and reviewing. Through iterative processes, this methodology improves the system to meet organizational needs while focusing on flexibility and user-centered development.

In conclusion, both articles use different approaches even though their goal is to improve cooperative information systems. The OSSAD framework presented in the first article offers a formal, model-driven method that prioritizes thorough analysis and organized design. The second article, on the other hand, combines Agile methods with traditional research, giving iterative development and ongoing user feedback priority for flexibility. Despite these variations, all approaches emphasize communication and process optimization, organizational analysis, and an organized method to data collection.

2.4 Summary

In conclusion, This chapter delves into the critical role that Information Systems (IS) play within organizations, shedding light on their significance in facilitating efficient operations and decision-making processes. It elaborates on the necessity of employing structured analysis and design methodologies to ensure the development of effective IS solutions. Furthermore, the document explores the far-reaching impact of IS on various aspects of organizational performance, underscoring the importance of aligning IS strategies with overarching business objectives to enhance competitiveness and drive success in dynamic market environments.

Chapter 3: Methodology

3.1 Introduction

In this chapter, the focus is on the process of system analysis and design, particularly project planning, a crucial phase for achieving successful outcomes. This chapter covers everything from feasibility studies that assess operational, technical, and economic factors to managing project schedules using tools like Work Breakdown Structures (WBS), Gantt charts, and PERT diagrams. It also provides practical insights and guidance. As we go through the System Development Life Cycle (SDLC), this chapter explains the roles of system analysts, highlighting the importance of collaboration, requirements gathering, and solution design.

3.2 The chosen methodology

The waterfall methodology definition places an emphasis on a step-through-step improvement in which each movement logically follows its predecessor to guarantee that every level is completed before moving directly to the subsequent. While there is lots of room for reviewing stages in response to issues that are determined or requirements that trade, tremendous looping is notion to be unusual and suggestive of beyond shortcomings. Making crucial choices at the right points in the lifestyle cycle, specifically early on within the technique of creating specific gadget requirements, is a crucial aspect of this linear method. Though viable, sizable modifications in later levels need to be strictly regulated and discouraged. The linear technique places a sturdy emphasis on early-stage studies, and whilst moderate changes made after deployment are permitted, they should not require a complete redesign.

3.3 Phases of the chosen methodology

The chosen methodology is the waterfall methodology. This chosen methodology consists of 6 phases which are planning, analysis, designing, implementation, testing, and maintenance but for this semester, we will conduct the first 3 phases only. Which are planning, analysis and design. Firstly, the planning phase is about getting information and collecting data for the project needs through conducting interviews, questionnaires, and brainstorming sessions. By following our project, planning phase is where we conducted an interview session with KADA. For example, during the interview session, we asked them about the current system that they are using, what kind of problem that they are facing by using the current system, how the cooperation system works, and how they would like their current system to be improved on. We take note off all the answers because that is where the key point to the next phases for our project.

Secondly, is the analysis phase. In this phase, where we would break down the main problem into smaller subtasks and divide the works equally among the team members. In this phase also, we would conduct the feasibility study which are technical, operational, and economic. From here, we would get the rough view of how much cost needed for this project, what position is needed for the improved project, and determining whether the system will function and be used after it is implemented and heavily depends on the human resources available for the project or not. That is what basically the phases is functioning.

Thirdly, in phase designing is about putting our initial decisions into practice. In this phase, we would record the hardware requirements and the programming language we plan to use if we want to develop a system software project. For our project, for the time being, we just used the high fidelity prototype, which we will be designing our system in Figma. From here, we would evaluate whether the system is meet the client requirement or not.

Fourthly, the implementation phase is part where we put our planned and documented activities into practice. This is the phase where we will spend most of our time. Especially if we want to develop a software system, this phase will mostly involve coding and meeting release and product development milestones.

Fifthly, this testing phase is where we search for issues with the implementation phase's deliverables. For example, if there is a mistake in the software code, or there are problems with the construction of the roof.

Lastly, the maintenance phase is where we tie up any loose ends, including small product improvements to enhance performance or address any problems or flaws.

3.4 Project planning schedule

3.4.1 Finding from feasibility study

1) Technical

After analyzing the current system, where the applicant has to get and submit the form manually, we found that KADA has the potential to develop an online system. This is because currently all the work is done manually and there are a lot of problems occurring. Therefore, by having an online system, it could increase the work efficiency and the number of customers. To make this a success, KADA has to hire a staff member in order to build and maintain the system. For example, KADA could hire an administrator that is a programmer where he can be a developer, and at the same time he also could be a system tester and expert.

2) Operational

Operational feasibility involves determining whether the system will function and be used after it is implemented and heavily depends on the human resources available for the project. This is why we have to ask the users and our client opinion regarding the current system, whether it needs an improvement or not. We also have to take into consideration the staff condition whether they are okay with the changes that want to be made or not. From the interview with our client, KADA in the past few days, our client wants to make an improvement by making their registration form digitally so that KADA could work more efficiently and increase the number of users in applying for membership and loan.

3) Economic

In order to make the system, we have to take into consideration the economic feasibility such as time, cost of doing a full system, cost of business employee time, and estimated cost of software, or software development. To have a clarity of the cost calculation, this is where we conducted a Cost-Benefit Analysis (CBA). By having the CBA, we could analyze and estimate whether the investment is worth it or not for a long term. If it is worth it, then the system is economically feasible, and the project should be proceeding further.

3.4.2 Cost benefit analysis(CBA)

The purpose of the cost-benefit analysis is to compare cost and benefit of a project or action in order to assist companies in making better decisions. Our main objective in conducting the CBA is to identify the profitability index (PI). To determine the project's PI, project's initial investment amount is divided by the present value of future estimated cash flows. If the PI is greater than 1.0, it means that it is considered to be a good investment.

Cost	
Hardware	20 000
Software	15 000
Training	10 000
Consultation	10 000
IS Support	10 000
Supplies	1200
Maintenance	2000
Advertisement	1250

Benefits	
Savings	40 000
Increase sales	20 000

Discount rate	15%
Sensitivity factor (cost)	1.1
Sensitivity factor (benefit)	0.9
Annual change in cost	5%
Annual change in benefit	7%

Cost	Year 0	Year 1	Year 2	Year 3
Development cost				
Hardware	22 000			
Software	16 500			
Training	11 000			
Consultation	11 000			
Total development cost	**60 500**			
Production cost				
IS Support		11 000	11 550	12 128
Supplies		1320	1386	1455
Maintenance		2200	2310	2426
Advertisement		1375	1444	1516
Total production cost		15 895	16 690	17 525
Present value		13 821	12 620	11 523
Accumulated cost		**74 321**	**86 941**	**98 464**
Benefit				
Saving		36 000	38 520	41 216
Increase sale		18 000	19 260	20 608
Total benefit		54 000	57 780	61 824
Accumulated benefit		**54 000**	**111 780**	**173 604**
Gain or lose (npv)		**(20 321)**	**24 893**	**75 140**
Profitability index	**1.24**			

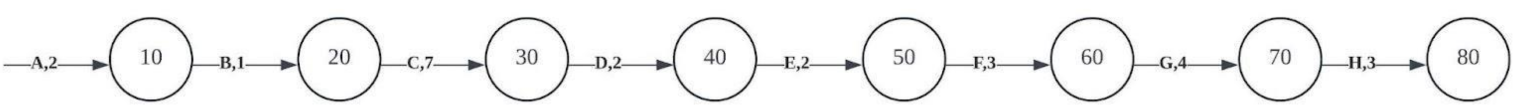
Therefore, it shows that this project would give a good investment because the PI is above 1.0.

3.4.3 Work based structure

Work based structure is a project management tool that divides a project's scope into manageable sections and organizes the entire project. When the work has been divided into sections, then the works also could be distributed to all the team members equally. Refer Appendix A which shows the work based structure for the system.

3.4.4 Pert chart

Task	Activity	Predecessor	Duration (days)
A	Interviewing user	none	2
B	Summarizing knowledge obtain	A	1
C	Generate proposal	B	7.
D	Create data flow diagram	C	2.
E	Analyze the structure decision made	D	2
F	Design procedure for data entry	E	3
G	Design interface	F	4
H	Design backup procedure	G	3



3.4.5 Gantt chart

KADA project

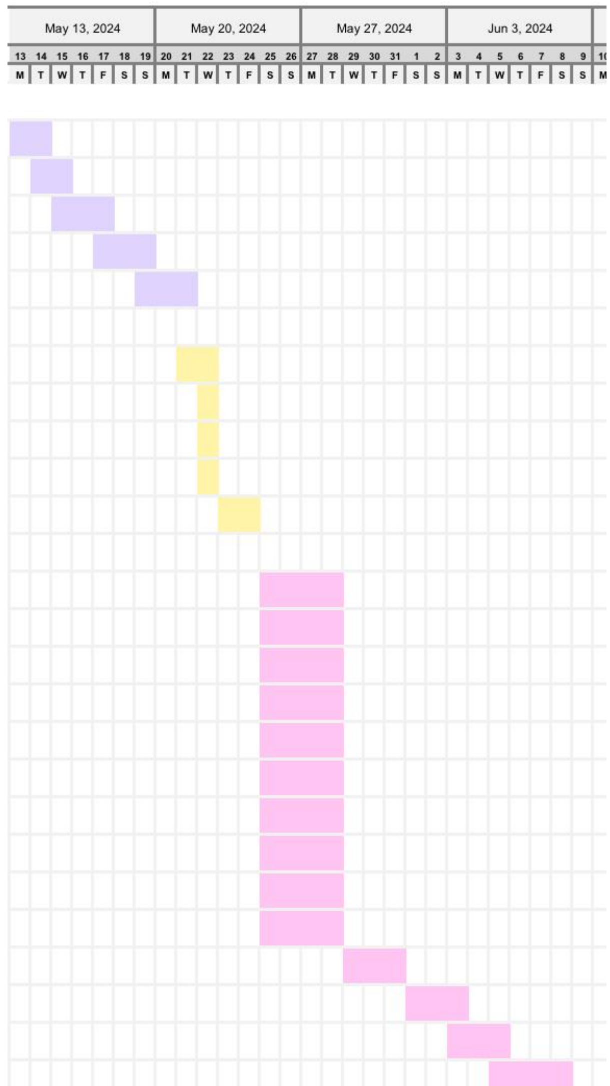
Muhammad Mukhritz **Project Manager**

Project start: **Mon, 5/13/2024**

Display week: **1**

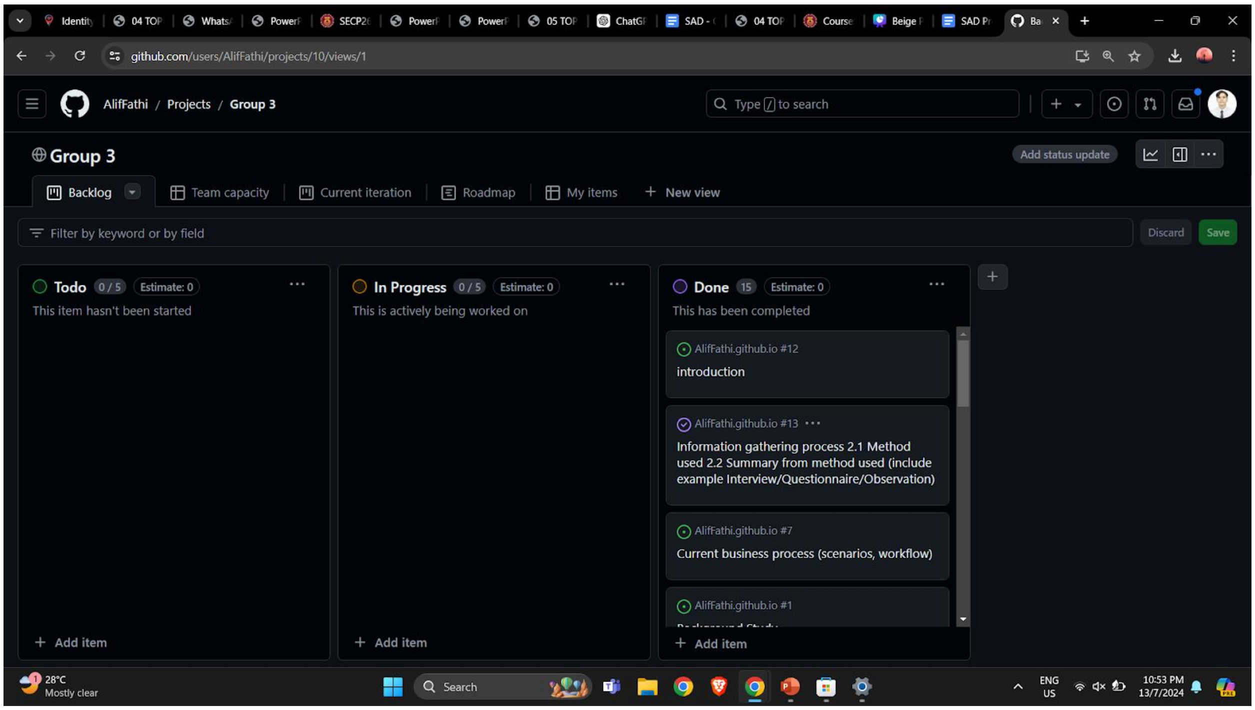
SIMPLE GANTT CHART by Vertex42.com
https://www.vertex42.com/ExcelTemplates/simple-gantt-chart.html

TASK	ASSIGNED TO	PROGRESS	START	END
Planning				
Conduct an interview with the representative of KADA	Naim, Yasmin	100%	5/13/24	5/14/24
Determine the scope of study	Mukhritz, Dania	100%	5/14/24	5/15/24
Determine the user requirements for the purpose study	Dania, Yasmin	100%	5/15/24	5/17/24
Choose methodology for the project	Dania, Yasmin	100%	5/17/24	5/19/24
Divide the task to be done	Dania, Yasmin, Mukhı	100%	5/19/24	5/21/24
Analysis				
Extract the key point from the interview	Mukhritz, Naim	100%	5/21/24	5/22/24
Determine the problem statements	Dania, Yasmin	100%	5/22/24	5/22/24
Determine the proposed solution	Mukhritz, Dania	100%	5/22/24	5/22/24
Determine the project objective	Naim, Yasmin	100%	5/22/24	5/22/24
Make comparisson between old system and proposed system	Mukhritz, Naim, Dania, Yasmin	100%	5/23/24	5/24/24
Design				
Ilustrate the logical (As-Is) DFD	Naim, Yasmin, Dania, Mukhritz	100%	5/25/24	5/28/24
Process 1 : Login and registration	Naim	100%	5/25/24	5/28/24
Process 2 : Appiving for cooperative membership	Naim	100%	5/25/24	5/28/24
Process 3 : verifying membership application	Naim	100%	5/25/24	5/28/24
Process 4 : Making a loan	Dania	100%	5/25/24	5/28/24
Process 5 : Checking and verifying loan applications	Dania	100%	5/25/24	5/28/24
Process 6 : Making a financial calculation	Yasmin	100%	5/25/24	5/28/24
Process 7 : Inform loan status to the applicant	Yasmin	100%	5/25/24	5/28/24
Process 8 : Making a report	Mukhritz	100%	5/25/24	5/28/24
Process 9 : Entering and maintaining data in the system	Mukhritz	100%	5/25/24	5/28/24
Ilustrate the logical (To-Be) DFD	Mukhritz, Naim, Dania, Yasmin	100%	5/29/24	5/31/24
Ilustrate the physical (To-Be) DFD	Mukhritz, Naim, Dania, Yasmin	100%	6/1/24	6/3/24
Create a wireframe for the user interface	Mukhritz, Naim, Dania, Yasmin	100%	6/3/24	6/5/24
Transform the witeframe to the figma	Mukhritz, Naim, Dania, Yasmin	100%	6/5/24	6/8/24



3.4.6 Github

We are using GitHub for team collaboration to effectively track each member's progress by monitoring the backlog items assigned to them. This approach allows us to manage tasks, monitor workflow, and ensure everyone stays on track with their responsibilities. By utilizing GitHub's features, we can maintain clear visibility on each team member's contributions and progress.



3.4 Summary

To conclude, this chapter provides an overview of the project planning process for system analysis and design projects. It covers key aspects such as feasibility studies, project scheduling techniques, and the roles of system analysts. Feasibility studies involve operational, technical, and economic assessments, including cost-benefit analysis. Project scheduling methods include work breakdown structures (WBS), Gantt charts, and PERT diagrams. Effective project management ensures successful outcomes by defining objectives, allocating resources, and managing timelines.

Chapter 4: System Analysis and Design

4.1 Introduction

The process of systems analysis and design involves a methodical approach to problem identification, opportunity analysis, and objective setting. It also involves studying the information flows inside businesses and building computerized information systems to address specific needs. The System Development Life Cycle (SDLC) is the most common methodology that people use in order to approach business solving problems. SDLC consists of 7 phases, which are identifying problems, determining human information requirements, analyzing system needs, designing the recommended system, developing and documenting system software, testing and maintaining the system, and implementing and evaluating the system.

4.2 System Gathering Technique

4.2.1 Method used

We have determined that the interactive way of information collecting for this project will be conducting interviews. This approach works well since it allows us to get the essential data for the system requirements. In order to avoid asking the interviewee the same questions twice, we have worked with other groups to create a set of questions in advance of the interview.

There are 2 types of questions which are open-ended questions and close-ended questions. The purpose of open-ended questions is to learn about the primary preferences and system objectives of the clients. Respondents are able to provide specific details about their requirements and expectations by answering these questions. Closed questions are created to gather precise data, trustworthy information and help us make informed decisions with clear data. By combining the two kinds of inquiries, we are able to fully comprehend the needs of our consumers. This is the example of the question :

- Bolehkah pihak tuan memberikan penjelasan mengenai sistem yang sedia ada?
- Apakah kriteria khusus yang diambil kira oleh pihak koperasi dalam proses kelulusan pinjaman, dan berapa lama biasanya masa yang diambil dari permohonan sehingga kelulusan pinjaman diberikan?
- Bahagian manakah yang tuan rasa perlu dikekalkan seperti yang sedia ada?
- Jika permohonan untuk menjadi ahli koperasi tidak diluluskan, adakah boleh untuk membuat rayuan semula?
- Adakah terdapat jumlah maksimum pinjaman yang boleh kita pinjam?

Following our preparation, on May 14, 2024, we conducted an online interview with our client, Encik. Ahmad Rohailan Bin Hani dan Puan. Nor Zamani Binti Ismail from Lembaga Kemajuan Pertanian Kemubu. During this conversation, we were able to discover more about the system requirements of the customer. The customer also supplied multiple comprehensive explanations of the system requirements. Lastly, the client discussed their needs for the proposed system as well as the functioning of the current system.

4.2.2 Summary from method used

Appendix B showcases a comprehensive overview of the KADA organization membership registration form. Aspiring members must complete this form to join the esteemed organization. The registration process begins with applicants providing their personal details, including their full name, residential address, identification number, and other relevant information. Following this, applicants are required to furnish details about their family. The final section of the form is reserved for KADA's administrative use, where the organization will document the membership fee, which is set at 50 Ringgit per month.

Refer to Appendices C and D to see the overview of the loan application form. The applicant must be a staff of the KADA In order to apply for a loan. The applicant needs to fill in the loan requirement such as the type of loan and how much the applicant wants to borrow. Next, the applicant needs to fill in the personal information and also the applicant's statement. After that, the applicant needs to have 2 guarantors in order to apply for this loan. Lastly , the part of the verification from the employer and the office use will be filled by the KADA staff.

Appendix E showcases a detailed example of the Monthly Cooperative Member Financial Report. This comprehensive report provides vital information regarding the applicant's stock holdings, including share capital and other relevant financial data. It also outlines the details of any loans the applicant has taken, categorized by the type of loan, such as Al-Bai, Al-Innah, among others. The final section of the report is dedicated to the applicant's verification of the financial statement, ensuring accuracy and transparency in the financial reporting process.

Appendix F presents the loan repayment schedule in a detailed tabular format. This table illustrates a fixed profit rate of 4.20% and offers repayment durations ranging from 1 to 6 years. Each year corresponds to a specific monthly installment amount. For instance, if the loan amount is RM 1,000 with a repayment period of 1 year, the applicant is required to pay RM 86.83 per month, in addition to the RM 50 monthly membership fee. This schedule provides a clear and structured overview of the repayment obligations for the applicants.

4.2.3 Current business process (scenarios, workflow)

Flow charts are an essential process to do. This is because by doing a flow chart, we could identify the main process and simultaneously see the bigger picture of how the system is working. The Appendices G and H shows a flowchart on how the membership registration and applying a loan process in KADA is currently working.

4.2.4 Company (Cooperative Of KADA)

KADA (Lembaga Kemajuan Pertanian Kemubu) is an agriculture organization in Kelantan. The workers under KADA are being given the opportunity to become cooperative members of KADA and make a loan under KADA too. But, the current system, or the as-is system in KADA to provide such services is being done manually and there is where we saw that it has very low efficiency and productivity.

4.2.5 Logical DFD AS-IS system

Data Flow Diagram (DFD) is a graphical representation of data flow in a system. It can show the flow of data entering, leaving, and being stored. DFD could be divided into different levels, where it can offer different levels of system information. There are 3 levels of DFD that we are currently doing, which are context diagram, diagram 0, and child diagram.

4.2.5.1 CONTEXT DIAGRAM

A diagram which shows a system element in relation to the entities it interacts with. In the registration and loan application system diagram, we represented that with a blue box in the middle, and all of the function registering, and loan application entity was discovered, which are applicator, board member, and clerk. Refer Appendix I to see the context diagram.

4.2.5.2 DIAGRAM 0

Diagram 0 divides the main processes shown in the context diagram in Data Flow Diagram (DFD) into smaller processes to give a more detailed view of the system. At the diagram 0 phase, every subprocess is represented as a distinct process. Each subprocess's related data stores and flows are also displayed. Appendix J shows a diagram 0 of the system**.**

4.3 System requirement

4.3.1 Functional requirement

User Registration:

- Users should be able to register for a KADA membership online by filling out and submitting a registration form.
- The system should validate the information entered (e.g., required fields, correct data formats) before submission.
- Users should receive an email confirmation upon successful registration.

Membership Management:

- Users should be able to view and update their membership details.
- Admins should be able to approve or reject membership applications.
- The system should notify users via email about the status of their membership application.

Loan Application:

- Members should be able to apply for a loan online by filling out and submitting a loan application form.
- The system should validate the information entered (e.g., required fields, correct data formats) before submission.
- Users should receive an email confirmation upon successful loan application submission.

Loan Management

- Admins should be able to review, approve, or reject loan applications.
- The system should provide a dashboard for admins to view pending, approved, and rejected loan applications.
- Users should be notified via email about the status of their loan application.

Document Upload:

- Users should be able to upload necessary documents (e.g., proof of identity, income statements) as part of the registration and loan application processes.
- The system should allow admins to view and download these documents for verification.

Notifications:

- The system should send email notifications for important events, such as registration confirmation, loan application status updates, and membership approvals.

Reporting:

- The system should generate reports on user registrations, loan applications, and other key metrics.
- Admins should be able to download these reports in common formats (e.g., PDF, Excel).

These functional requirements ensure that KADA's online system will support essential processes such as user registration, membership management, loan application processing, and administrative oversight, while providing a secure and user-friendly experience.

4.3.2 Non functional requirement

Performance:

- The system shall handle up to 2000 simultaneous users without significant performance issues.

Usability:

- The system will provide a user-friendly interface that is easy to navigate.

Security:

- The system will employ security measures such as firewall to prevent unauthorized access

Maintainability:

- The system shall ensure a straightforward system design that allows for easy updates and maintenance

Backup and Recovery:

- The system will implement regular backups of critical data.

4.4 System design

4.4.1 Proposed system: Logical DFD (TO BE) System (context diagram, diagram 0, Child diagram)

Context diagram TO BE

A context diagram is a diagram that shows a system element in relation to the entities it interacts with. In the registration and loan application system diagram, we represented that the box in the middle shows the system for registration and loan application system, and all of the function registering, and loan application entities for the system, which are the applicator, board member, and clerk. See Appendix K for the to-be context diagram.

Zero diagram TO BE

Diagram 0 divides the main processes shown in the context diagram in Data Flow Diagram (DFD) into smaller processes to give a more detailed view of the system. In the diagram 0 phase, every subprocess is represented as a distinct process. Each subprocess's related data stores and flows are also displayed. There are 9 processes in this membership application and loan registration for this system. The process for this system starts with login and registration followed by applying for the cooperative membership, verifying membership application, making a loan application, checking and verifying loan application, making financial calculations, informing loan status, making a report and ending with entering and maintaining data in the system. 4 entities are involved in this system which are applicator, clerk, ALK and admin. Appendix L shows a diagram 0 of the system.

Child Diagram TO BE

This child diagram is for the login and registration process. Firstly the applicant will fill in their personal data to sign up for their account. Next, They will get the email verification that shows their verification status. Then, they need to enter their username and password to log into the system. Figure 4.4.1.1 shows a child diagram of the system.

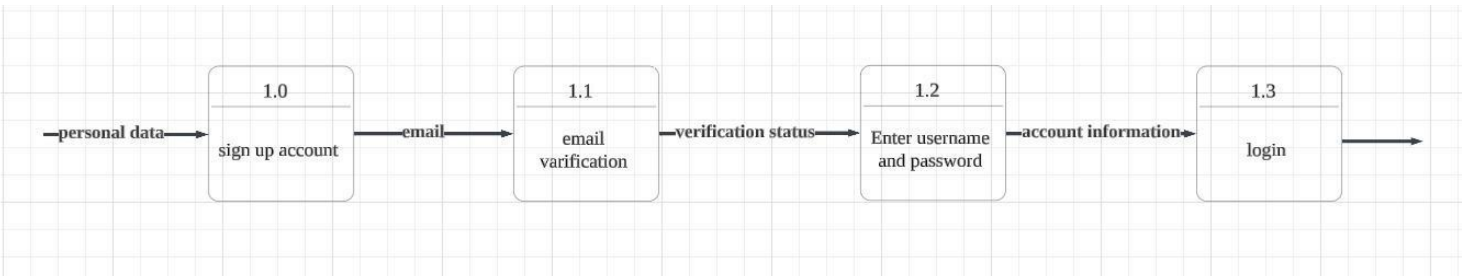


Figure 4.4.1.1

This child diagram is for applying for cooperative membership. Firstly the applicant will fill in their personal information into the applicant form. Next, they need to fill in family information in the family section of the form. Lastly, they need to fill in fees and contributions. The system will store the applicant personal information, family information and finance information. Figure 4.4.1.2 shows a child diagram of the system.

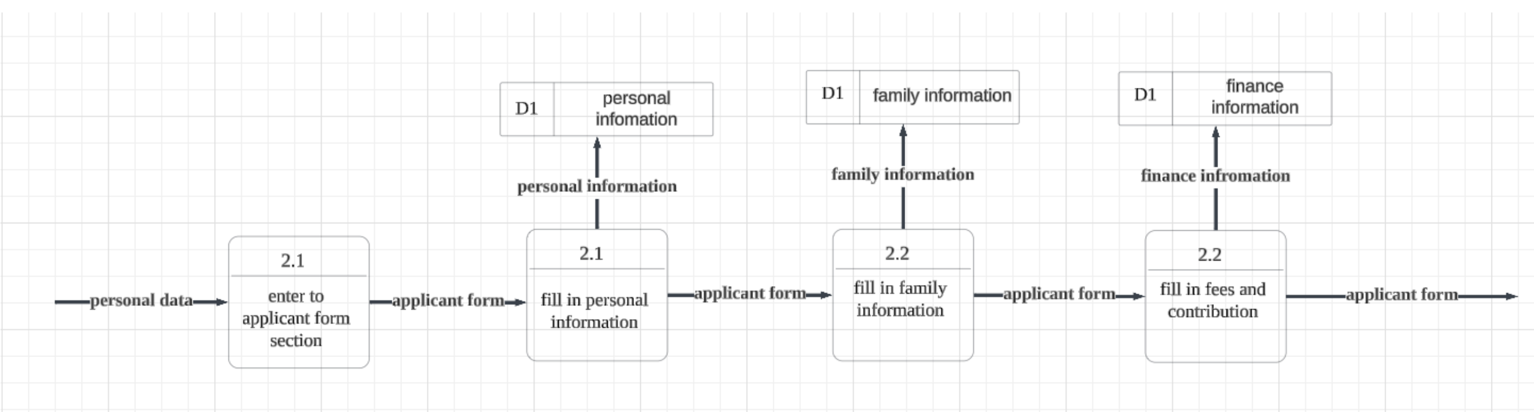


Figure 4.4.1.2

This diagram shows a process for the verifying membership application. Firstly, the system will get the personal data from the applicant. Next, the clerk will check the validity of the personal data and notify applicant status. Lastly, The system will generate digital cards for the successful application based on the info from the applicant personal data. The system will store the applicant personal information and family information. Figure 4.4.1.3 shows a child diagram of the system

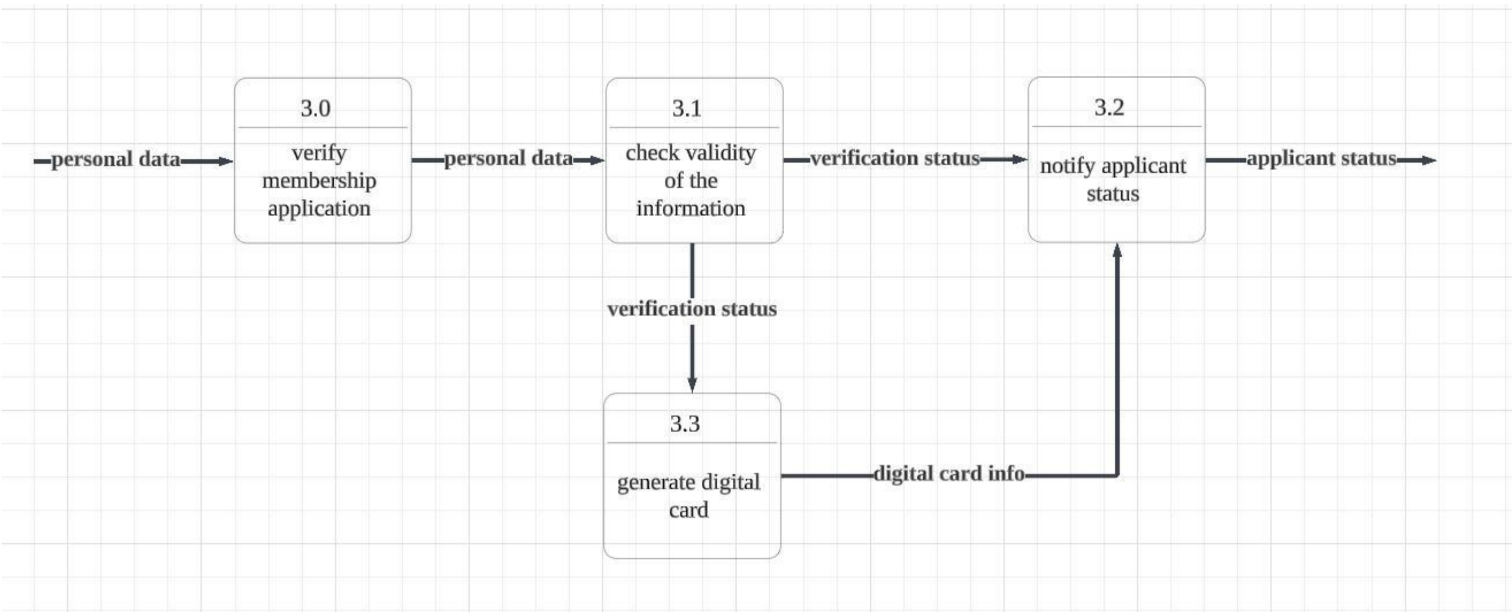


Figure 4.4.1.3

This child diagram shows the process for making a loan. Firstly, the applicant will choose the type of loan. The applicant will fill in their financing details followed by their personal information. Next, they need to fill in their confession form. After that, the applicant needs to fill in a guarantor form. Lastly, the applicant needs to upload the staff confirmation. The system will store the applicant personal information and family information. Figure 4.4.1.4 shows a child diagram of the system

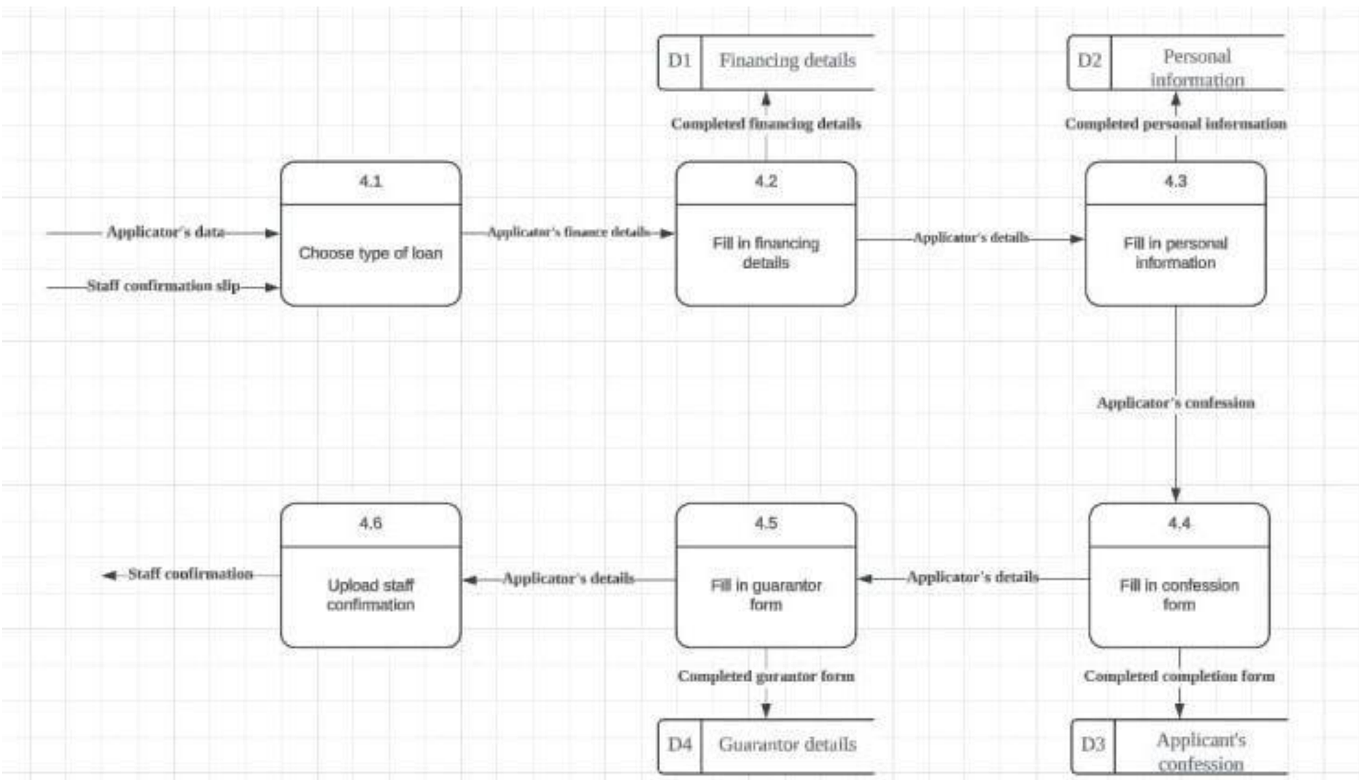


Figure 4.4.1.4

This child diagram shows a process for checking and verifying loan applications. Firstly, the clerk will inspect the information validity based on the application information. Lastly, the clerk will determine the status of the loan and notify the loan status to the applicant. The system will store the applicant personal information and family information. Figure 4.4.1.5 shows a child diagram of the system.

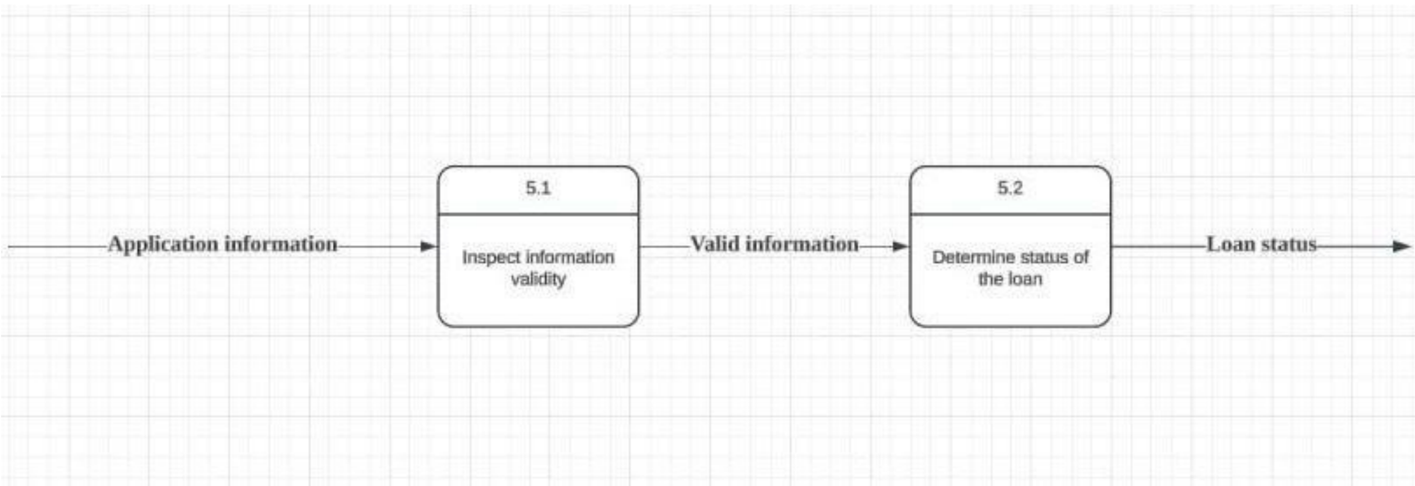


Figure 4.4.1.5

This child diagram shows a process for making a financial calculation. Firstly, the clerk will identify the type of loan based on the financial details from the user. Next, the clerk will determine the repayment method and proceed to calculate the loan repayment. Lastly, the system will generate the financial statement for the loan. The system will store the applicant personal information and family information. Figure 4.4.1.6 shows a child diagram of the system

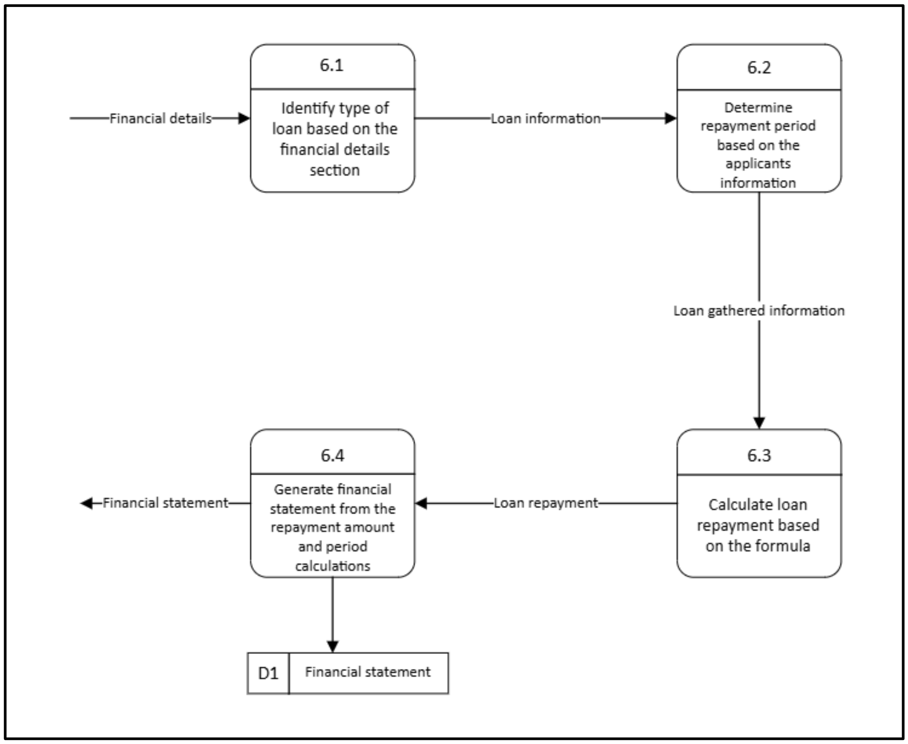


Figure 4.4.1.6

This child diagram shows a process to inform loan status to the applicant. Firstly, the clerk will compile the financial statement and loan statement of the applicant. Next, the status will be stored in the system. After that, the system will send an email to the applicants. Figure 4.4.1.7 shows a child diagram of the system.

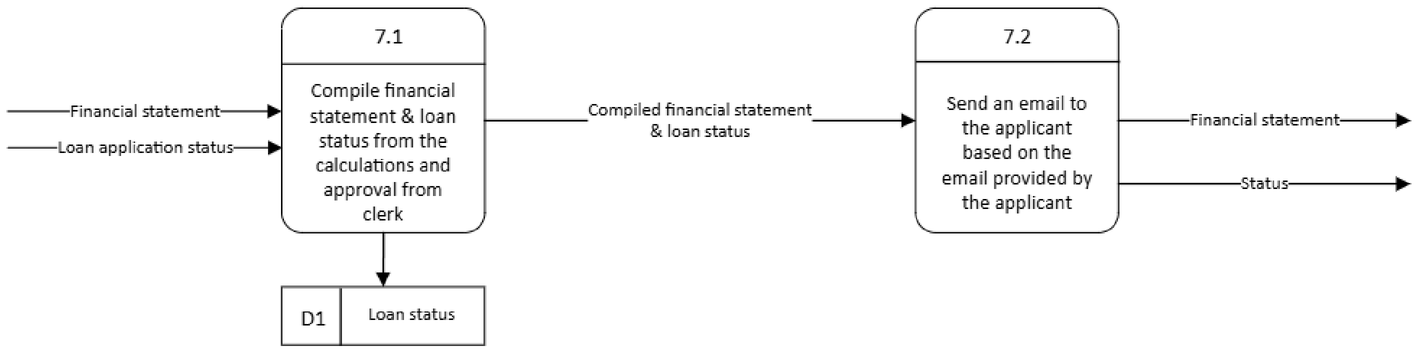
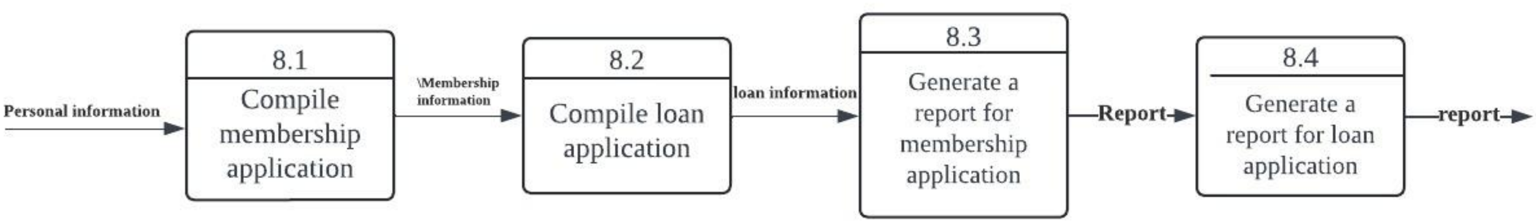


Figure 4.4.1.7

This child diagram shows the process of making a report. Firstly, the admin will compile membership applications based on the applicants personal information. Next, the admin will compile the loan application. After that, the system will generate a report for the membership application and also report for the loan application. Figure 4.4.1.8 shows a child diagram of the system.

Figure 4.4.1.8



This child diagram shows a process for entering and maintaining the data in the system. Firstly, the admin will schedule a regular backup for the system. Next, the admin will monitor system security for the system. After that, the admin will extract the information from the report and key in the information into the database. Figure 4.4.1.9 shows a child diagram of the system.

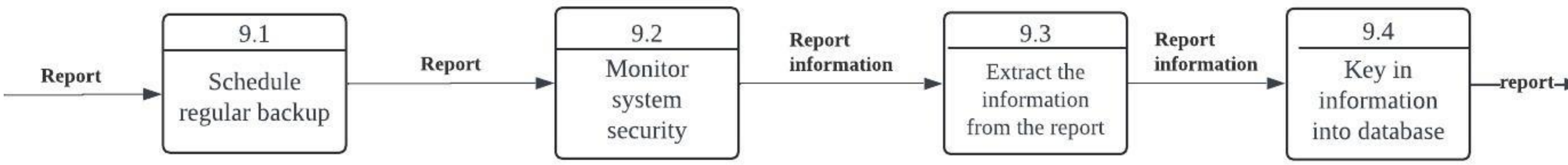


Figure 4.4.1.9

4.4.2 Proposed system: Physical DFD (TO BE) system

The physical diagram shows which processes are performed manually and which are automated. The processes are described in a more detailed manner in order to show the sequence of the process that must be done in a particular order and to identify temporary data stores.

Context diagram

A context diagram is an overview of the system that shows the basic inputs, general system, and outputs. This diagram is basically the bird's eye view of the data movement in the system. Appendix M shows our proposed system context diagram, with the entities being the applicator, board members, clerk, and admin.

Level zero

Level 0 physical diagram provides more detail by showing how the system will be constructed. It contains the manual processes, processes for adding, deleting, changing, and updating records, data entry and verifying processes. The processes involved in our proposed system are registering and key in username and password, key in accurate and complete details to apply for the cooperative membership, verifying membership application, key in financing details on the loan section, checking and verify loan application, key in data to the calculation template, send email to inform applicant's loan status, making a report and lastly entering and maintaining the data in the system. The entities involved are applicator, clerk, admin and ALK or the board members. Refer Appendix N to see the level zero diagram for the physical DFD (TO-BE) system.

Figure 4.4.2.1 shows the child diagram of the process of registering and keying in a username and password. The applicants will key in their data such as email, name and password to sign up for a KADA account on the website. They will then receive a verification code through their email to verify their email address. After verifying their email address, they can log in to the system.

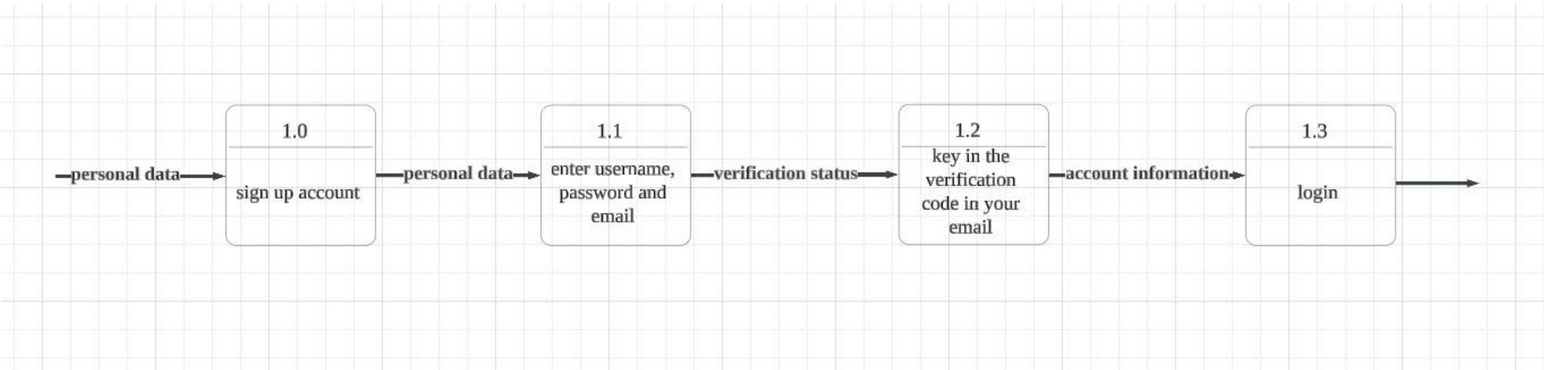


Figure 4.4.2.1

Figure 4.4.2.2 shows the child diagram of keying in accurate and complete detail to apply for the cooperative membership. To apply for KADA cooperative membership, the applicants must click “Permohonan menjadi anggota” and be directed to the registration tab. They will then fill in their personal information and family details. Both types of details will then be stored.

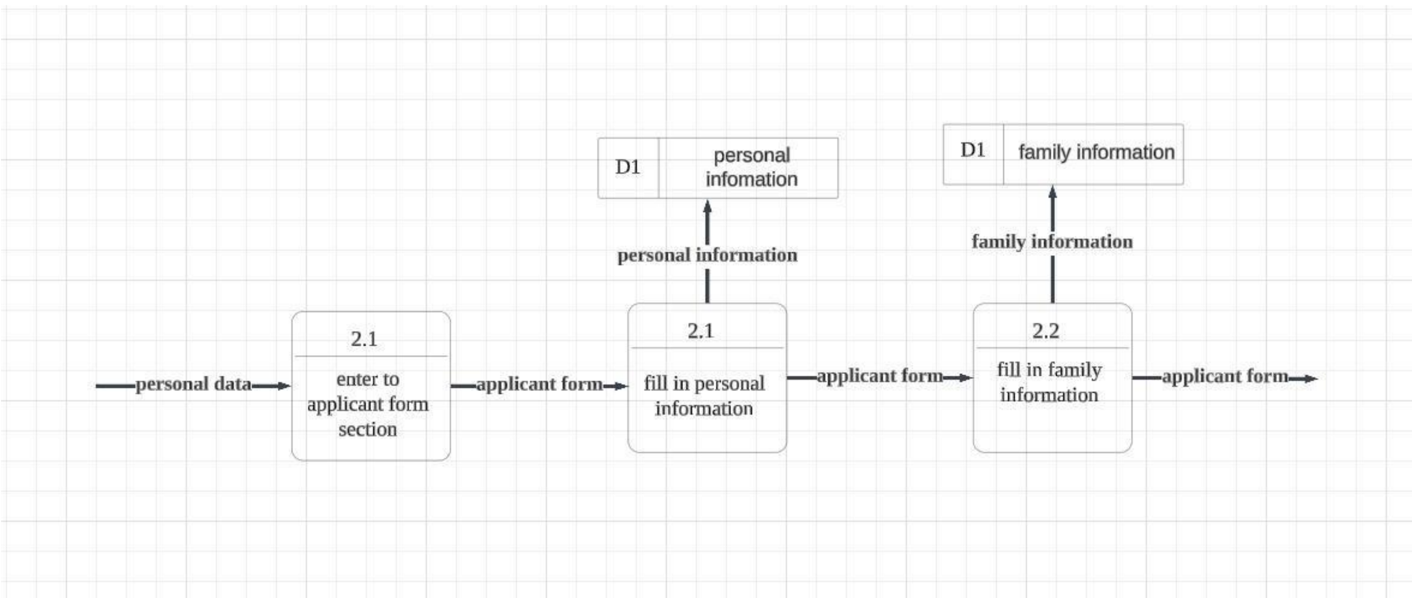


Figure 4.4.2.2

Figure 4.4.2.3 shows the child diagram of the process verifying the membership application which is done by the clerk. The clerk will click on the membership application tab and check the validity of the applicants' information. If the information given is valid, the clerk will filter the applicants who are eligible and also ineligible for the application. If the applicants' application is accepted, a digital card will be generated. All applicants will receive an email to inform them of their application status.

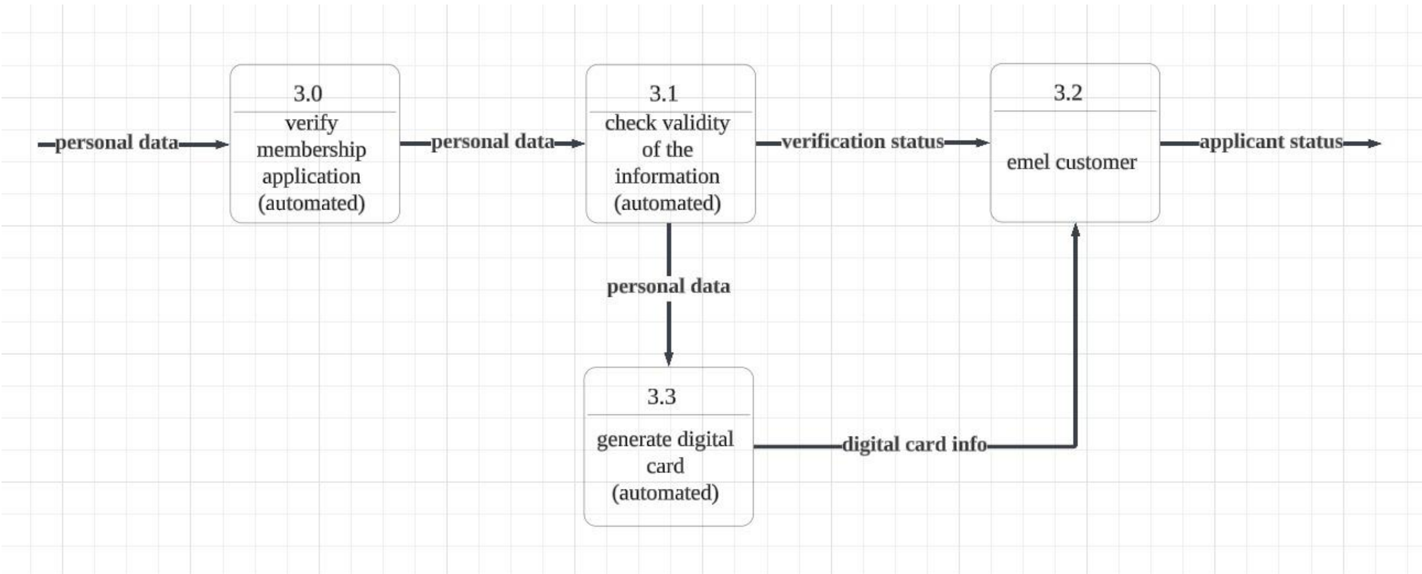


Figure 4.4.2.3

Figure 4.4.2.4 shows the child diagram of the process of keying in financing details on the loan section. Applicants who ought to apply for a loan with KADA will have to click on the loan application tab and choose the type of loan. Then, they need to fill in the required information such as their financing details, personal details and guarantor details. Next, they need to fill in the confession form in the personal details tab. Last but not least, they are required to upload their staff confirmation proof on the system.

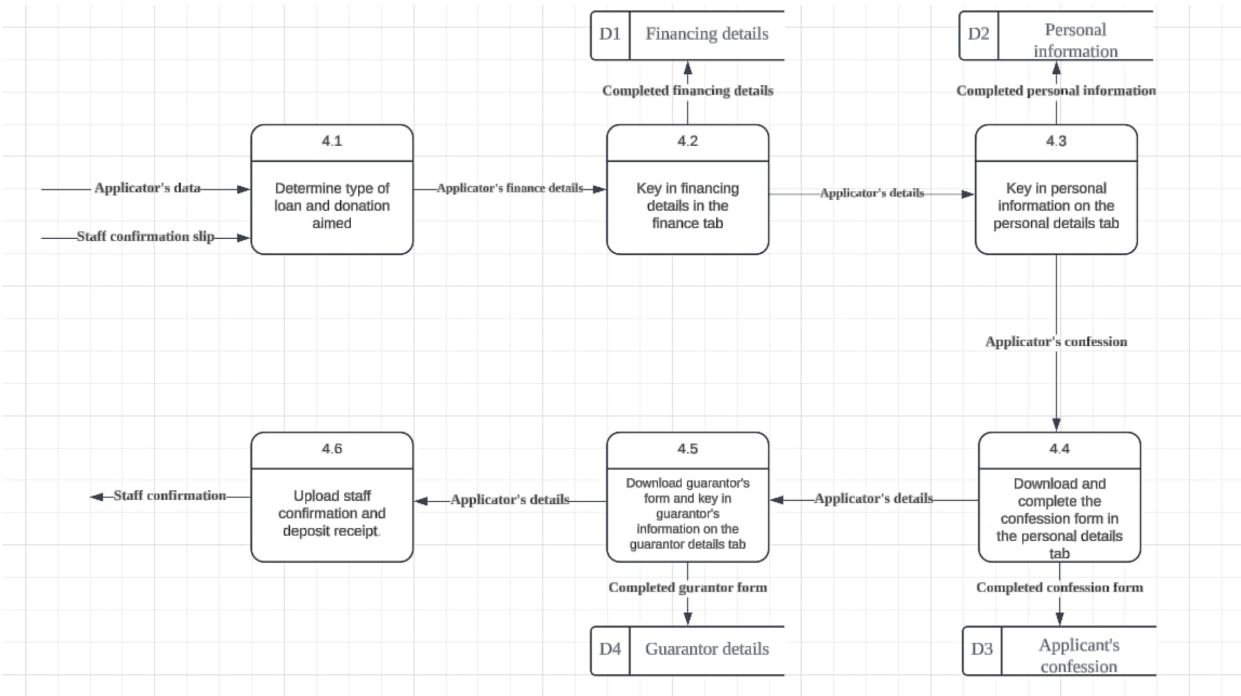


Figure 4.4.2.4

Figure 4.4.2.5 shows the child diagram of checking and verifying loan applications that are done by the clerk. The clerk will inspect the applicants' information and determine whether to accept or reject the loan application.

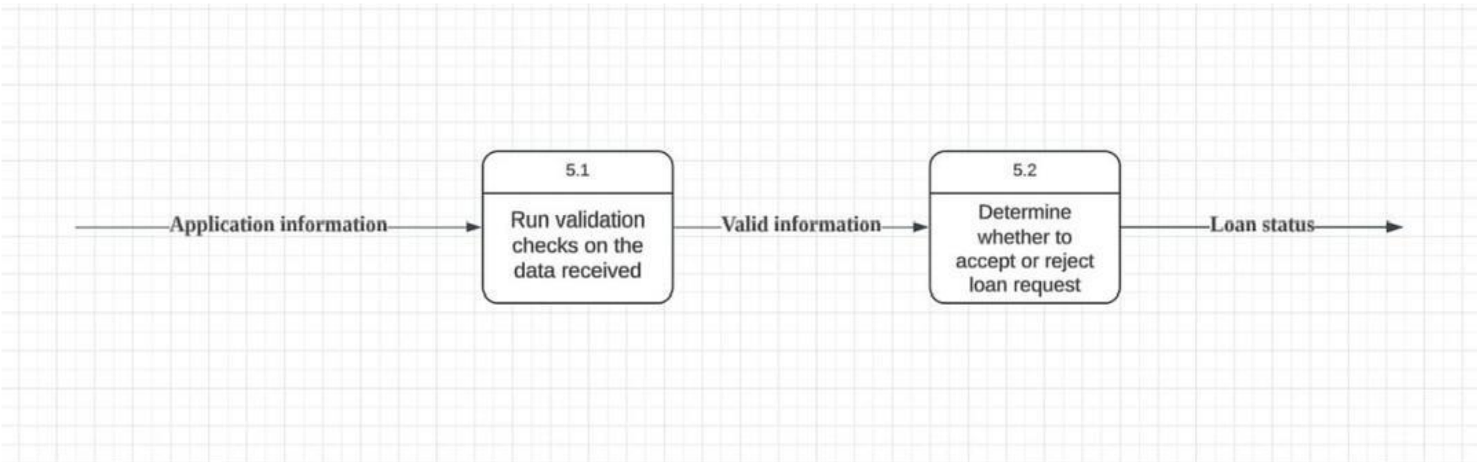


Figure 4.4.2.5

Figure 4.4.2.6 shows the child diagram of keying in data into the calculation template. This process is also done by the clerk. Clerk will first identify the type of loan chosen by the applicant based on the financial details section and determine the repayment period. Then, the loan repayment is calculated using the formula integrated with the system. Lastly, a financial statement will automatically be generated by the system and be stored.

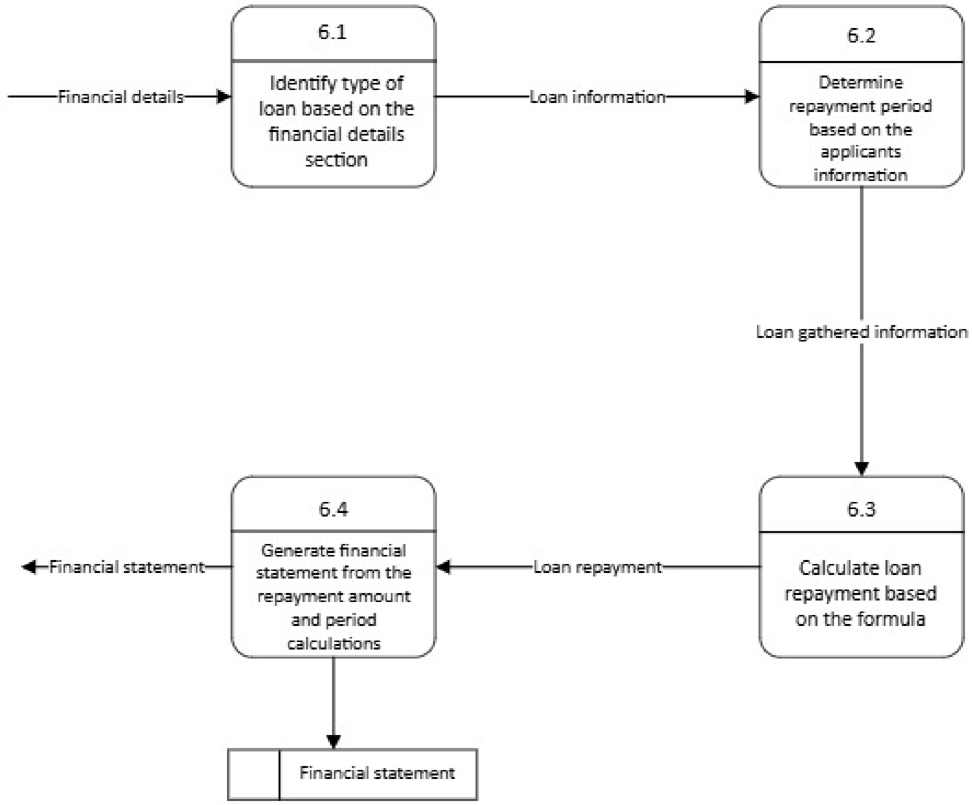


Figure 4.4.2.6

Figure 4.4.2.7 shows the child diagram of sending an email to inform the applicant's loan status. This process is automated. Financial statements, loan repayment plans and approval statements are compiled and sent to the applicant through the email provided by them.

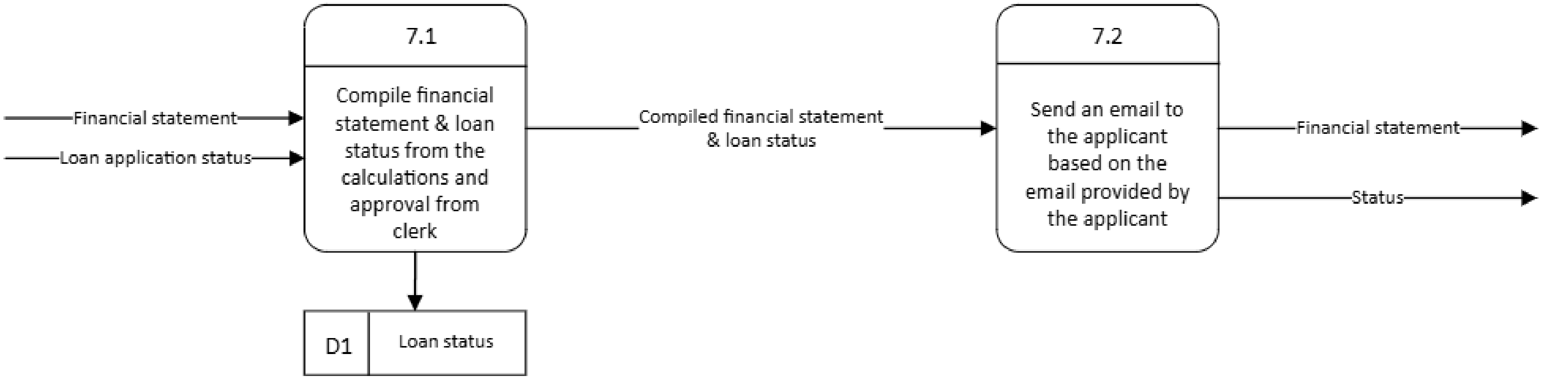


Figure 4.4.2.7

Figure 4.4.2.8 shows the process of making a report. This process is also another automated process in the system. All membership applications will be compiled, as well as all loan applications. Separate reports for both membership and loan applications will automatically be generated .

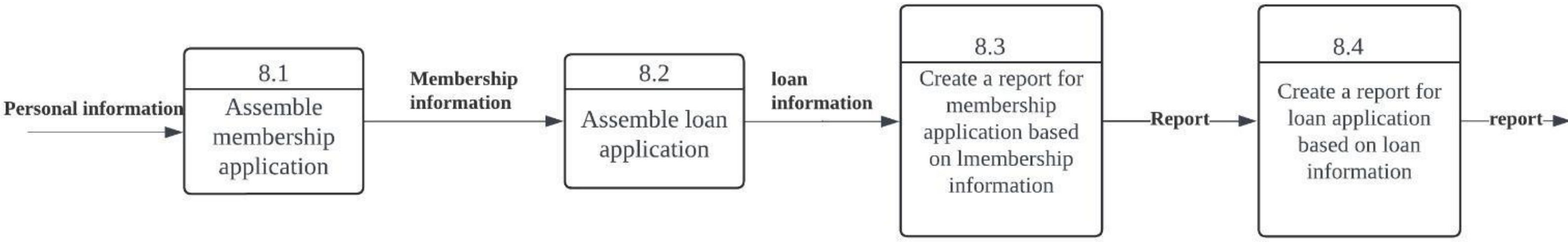


Figure 4.4.2.8

Figure 4.4.2.9 shows the child diagram of entering and maintaining data in the system. A regular backup is planned for the system, the admin will observe the system security, extract all information from the generated report and fill the information in the database.

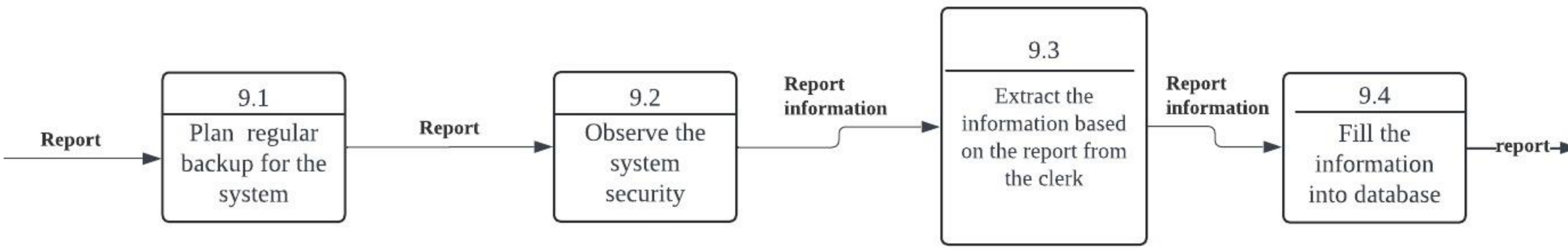


Figure 4.4.2.9

4.4.3 Process Specification

Process specification is a detailed description of the steps, procedures, and methodologies involved in completing a specific task or operation within a system or project. It outlines the precise sequence of actions, inputs, outputs, and the resources required to achieve the desired outcome.

Process 1.2: Key in the verification of code in your email

```
Begin
Enter verification code
If verification codes are correct
        Login to the website
Else
        Give an error statement
End if

Stop
```

3.0 Verify membership application

```
Begin
Verify membership application
IF the membership application is valid
        Check the validity of the information
        IF information are valid
                Generate digital card
        Else
        Send email to the applicator
Else
Send email to the applicator
End if
Stop
```

4.4.4 Partitioning

The provided Data Flow Diagram (DFD) is partitioned into distinct components based on user groups, timing, tasks, efficiency, data consistency, and security. The Applicant Component includes processes for user registration and application detail entry. The verification component, managed by clerks, handles membership and loan application verifications. The data maintenance and reporting component, involving both clerks and administrators, focuses on data maintenance and report generation. The financial details handling component ensures the secure processing of financial details and calculations. Finally, the notification component automates the process of informing applicants about their loan status. Each component is clearly demarcated to streamline operations and enhance security and manageability. Refer to Appendix O to see the DFD partitioning.

4.4.5 CRUD matrix

The CRUD Matrix is a technique for showing where each of these processes takes place in a system. The acronym of CRUD is used for Create, Read, Update, and Delete. Table 4.4.4.1 is CRUD Matrix for a new upcoming membership registering and applying a loan system in KADA. Some of the processes include more than one activity. For example, data entry processes such as keying and verifying.

Activity	Applicator	Clerk	Admin
Register and key in username and password	C		
Key in accurate and complete detail to apply for the cooperative member	C	R	
Verifying membership application		R, U	
Key in financing detail on loan section	C	R,	
Checking and verify loan application		R, U	
Key in data to the calculation template		C, U	
Send email to inform applicant's loan status		C	
Making a report		C	R
Entering and maintaining data to the system			C

Table 4.4.5.1

4.4.6 Event response table

Event response tables function as effectively record and monitor actions to potential events or situations. The input and output of the data flows is a trigger and response column, meanwhile the activity becomes the process. Analyzing the input and output of the data flows will allow the analyst to identify the data stores needed for the process. Table 4.4.4.1 illustrates the data flow diagram of the event response table.

Event	Source	Trigger	Activity	response	destination
Applicator register an account	Applicator	Applicator username and password	Create an account based on the applicator details. Send email verification.	Registration are complete	Applicator
Applicator login	Applicator	Applicator username and password	Find applicator record in system and verify password. Send to home page	Home page	Applicator
Applicator apply for member of cooperative KADA	Applicator	Applicator personal information.	Store the personal information. Send the registration form to clerk	Registration form	Clerk
Clerk verifying membership applicator	Clerk	Verification by Clerk	Check the details and verify the applicator information.	Status membership	applicator
Applicator apply for loan	Applicator	Applicator personal details and staff confirmation slip	Store the applicator personal details .send the loan form to Clerk. Send	Loan form	Clerk
Clerk verify and check the details of the applicator loan form	clerk	Verification by clerk	Check the details of the loan form and accept it if it meets the requirement. Send the loan status	Loan status	Clerk

Event	Source	Trigger	Activity	Response	Destination
Calculate loan repayment	Clerk	Applicator financial details and financial repayment period	Calculate the amount that applicator has to pay within the period of time that they provided.	Financial statement would be generated.	Clerk
Sending an email regarding to their loan status	Clerk	Applicator financial statement, loan status, and applicator details	Informing the status of the applicators loan statement through their email that provided in the applicator's details.	Results of the loan application	Applicator
Producing a company performance report	Clerk	Application of membership KADA and loan statement information	Making a performance report based on how efficient the system works as their data collection and room of improvement	Performance report could be observed	Admin and Ahli Lembaga Kada (ALK)
Updating and maintaining the data in the system	Admin	Report that has been generated and observed	To insert and keep maintaining the data from the report in the database as a reference for later use.	Data would be saved in the database	Database system

Table 4.4.6.1

4.4.7 Structure chart

The hierarchical organization of modules is represented by the organization. It breaks down a system into its most basic functional modules and provides a more thorough explanation of the main and supporting functions of each module. The control module refers to the main processes that are involved in the system. It is a higher level module that directs to the lower-level modules which are subordinate modules and possibly library modules. The arrow with an empty circle shows the data that is passed from one module to another.

Figure 4.4.5.1 shows a structure chart for the first process, which is a Login process. In the login process, if the user doesn't have an account, it would be directed to the Signup page. In the SignUp page, users have to create their username and password, and they have to provide their email address too. After that, they would receive a verification code through their email and they have to key in the verification code. Only then, they would have an account, and they would be able to login to the system.

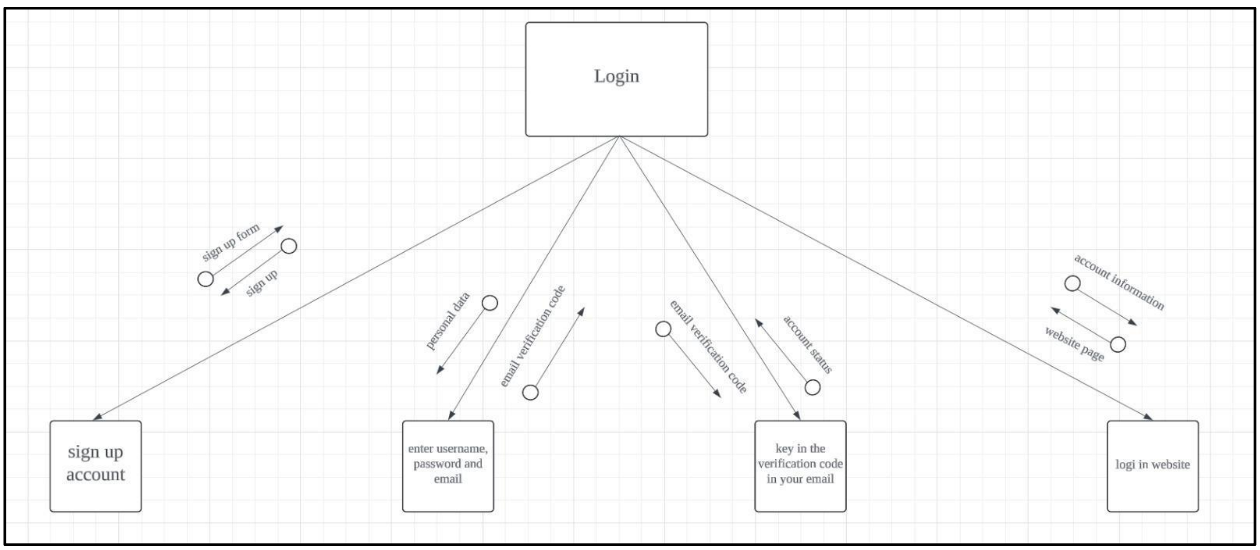


Figure 4.4.5.1

Figure 4.4.7.2 shows a key in details to apply for cooperative membership of KADA. In this process, the applicant would go to the applicant section form after the login process. In the form, the applicant would key in their personal information. And then, the form that has been fulfilled would be sent to the clerk to proceed to the next process.

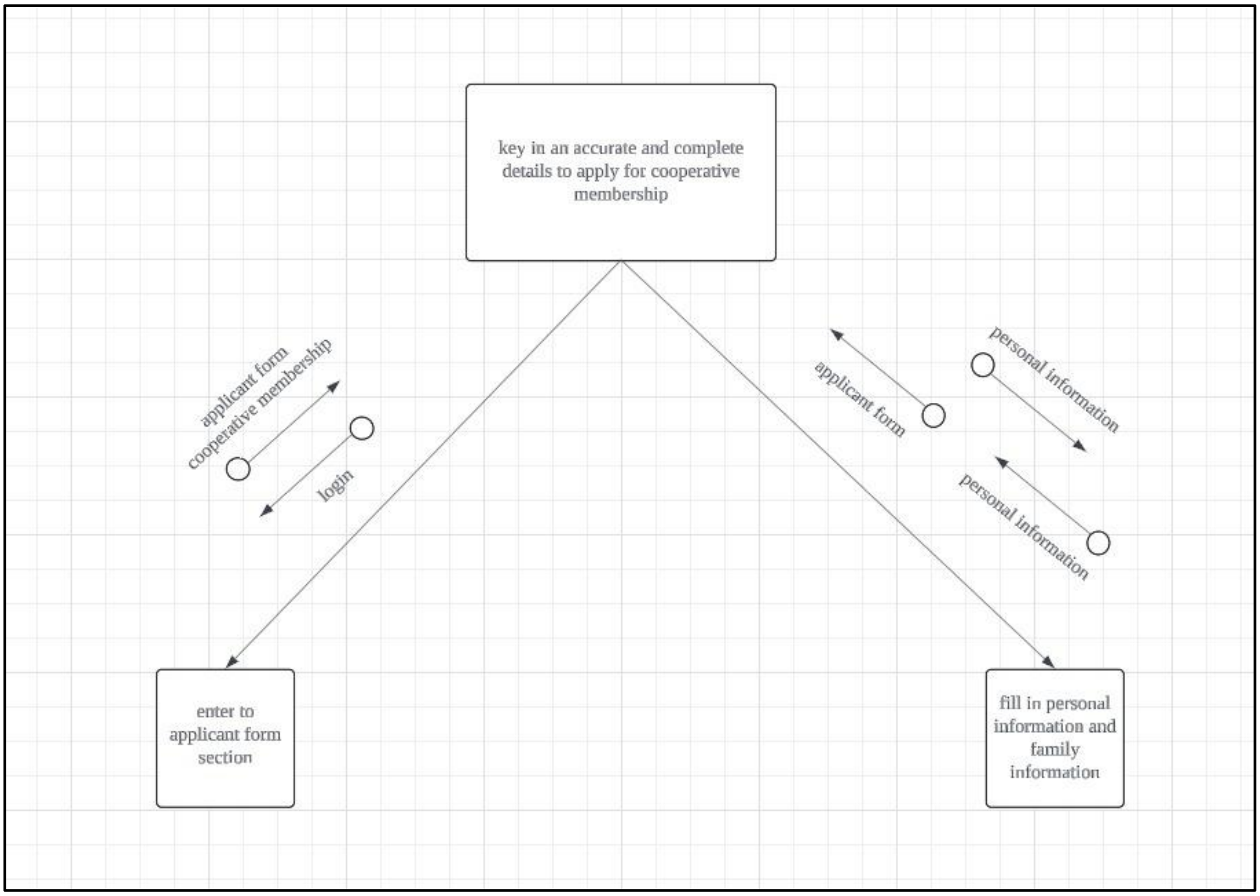


Figure 4.4.7.2

Figure 4.4.7.3 is a verifying membership application process. In this process, the clerk will be the one who would verify the membership based on the personal data and it would produce the status verification. Then, based on the personal data, it would check the validity of the information whether it's correct or not, then it would return the validity status. After the verifying process is done, the system would generate a digital card for the applicator use. Not only that, the system also would inform the applicator about their membership status through email.

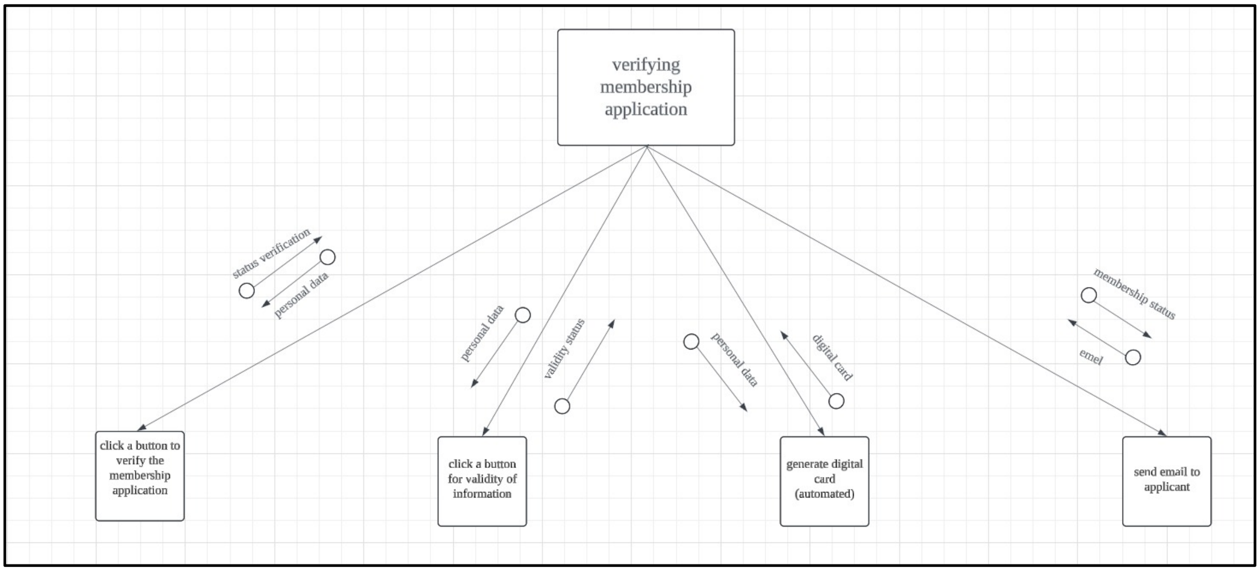


Figure 4.4.7.3

Figure 4.4.7.4 shows the structure chart for the process key in financing details on the loan section which is also one of the subordinate modules. In this module, the user will first have to select the type of loan and donation that they desire and key in their financing details in the system. Next, they will have to key in.

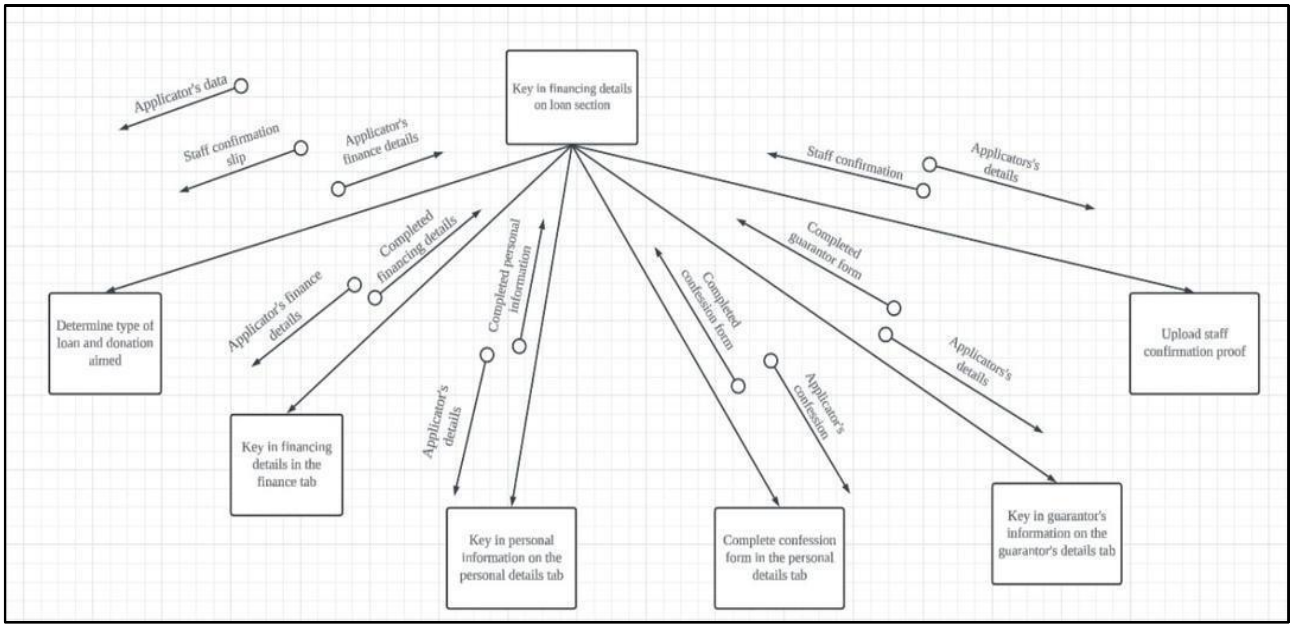


Figure 4.4.7.4

Figure 4.4.7.5 shows the structure chart for the process of checking and verifying loan applications. This process is done by the clerk. The clerk will run a validation check on the data received by the applicants. If the information is legitimate, the clerk will determine whether or not to accept the loan application. The determined loan status will then be handled in the next process.

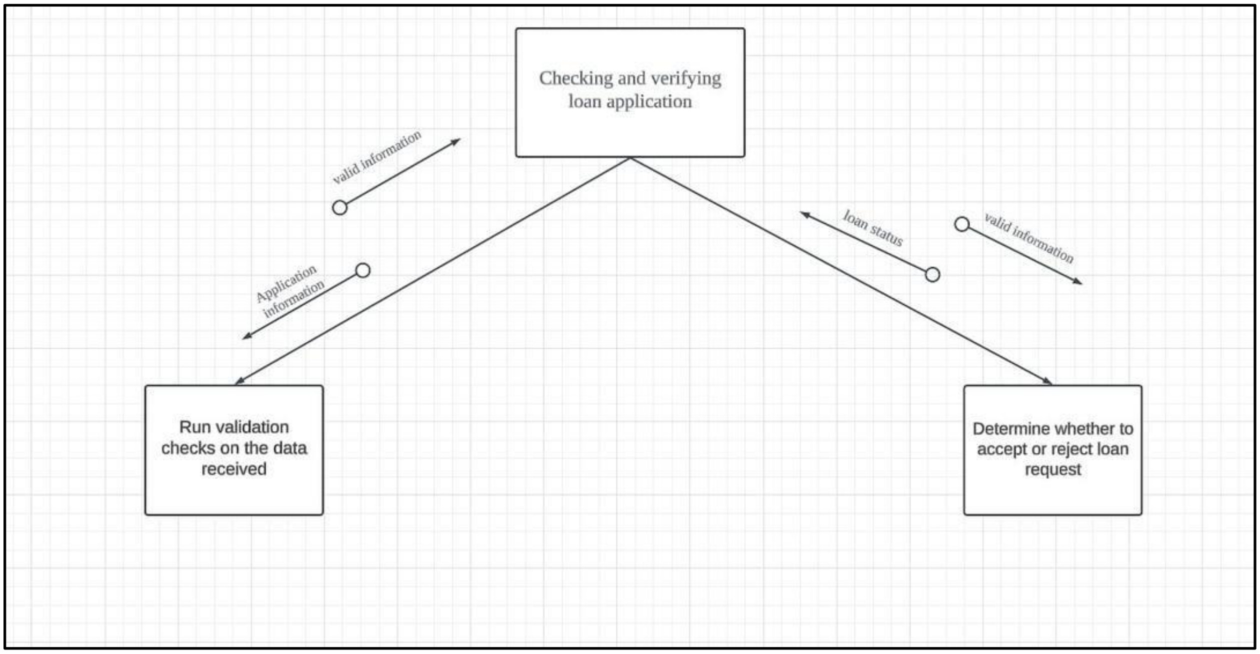


Figure 4.4.7.5

Figure 4.4.7.6 shows a key in data to the calculation template process. This process would be conducted by the clerk. To fulfill this process, firstly the clerk has to identify the type of loan. Then, it has to determine the repayment period. By having this information, then it can calculate the loan repayment based on the formula. After doing the calculations, it would generate the financial statement for the applicator use.

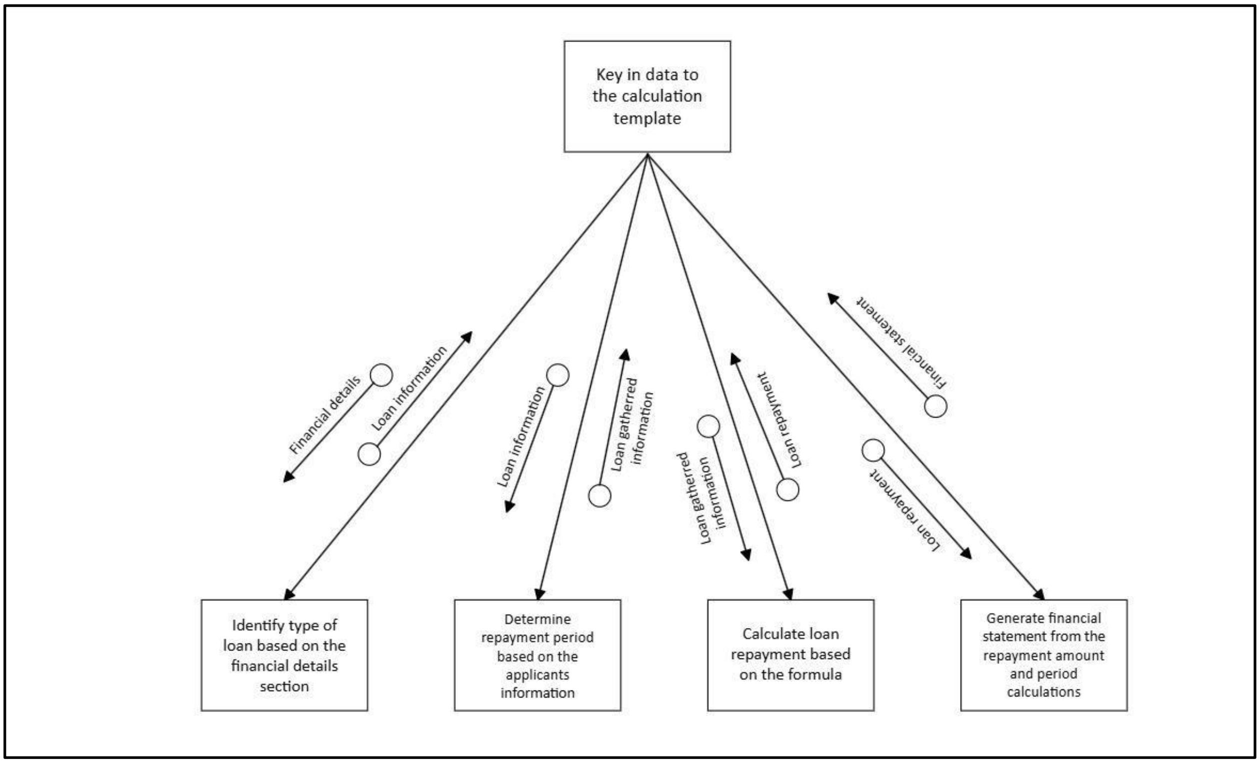


Figure 4.4.7.6

Figure 4.4.7.7 is a sending email to inform the applicator about their loan status. In order to fulfill this process, it has to receive the loan application status and financial statement. Then it would compile it, and send it through the applicator's email that is provided by the applicator.

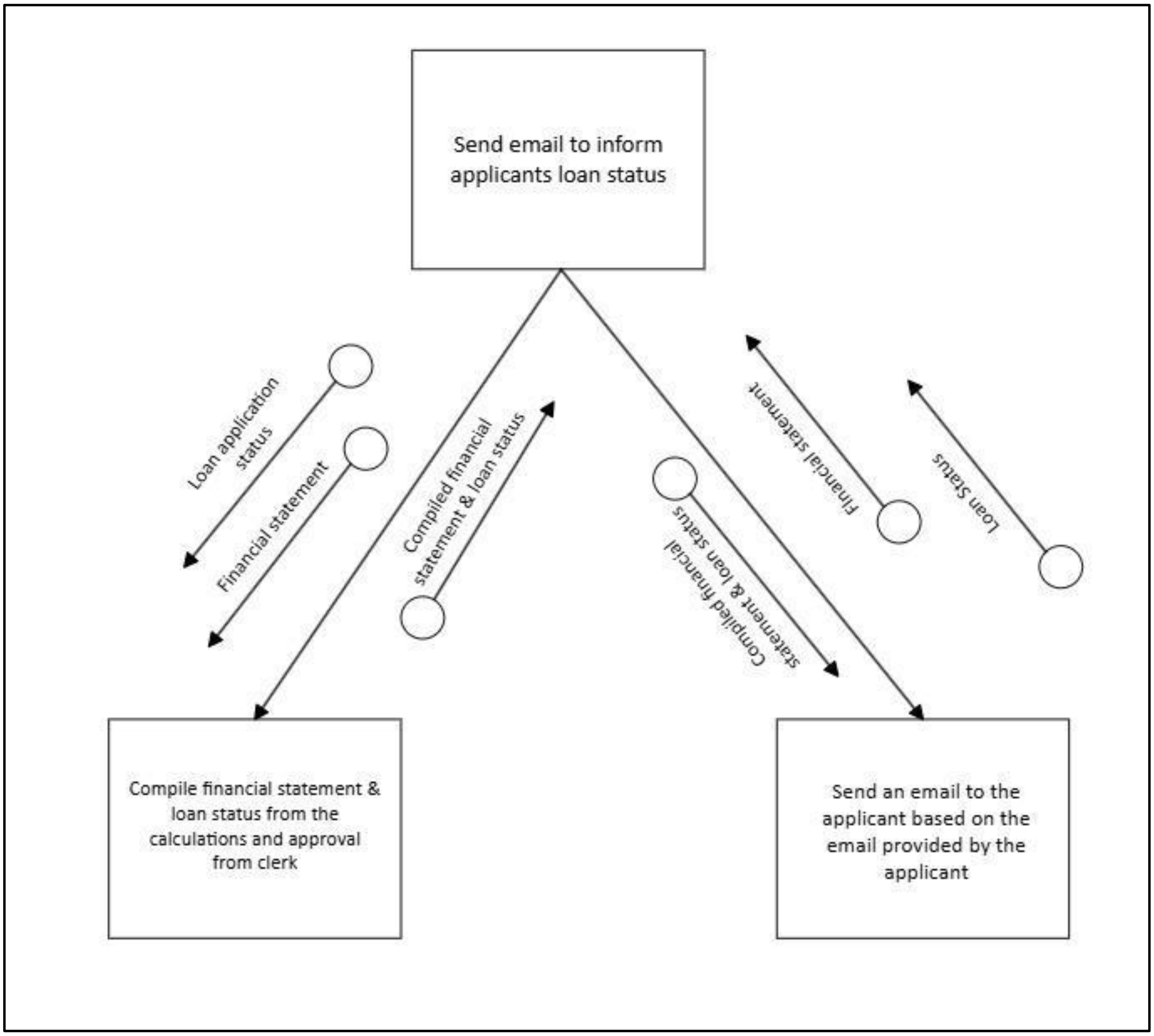


Figure 4.4.7.7

Figure 4.4.7.8 is a reporting process. To make a report, it has to put all together the membership application and loan application information. After gathering all the information, then only it would generate the report.

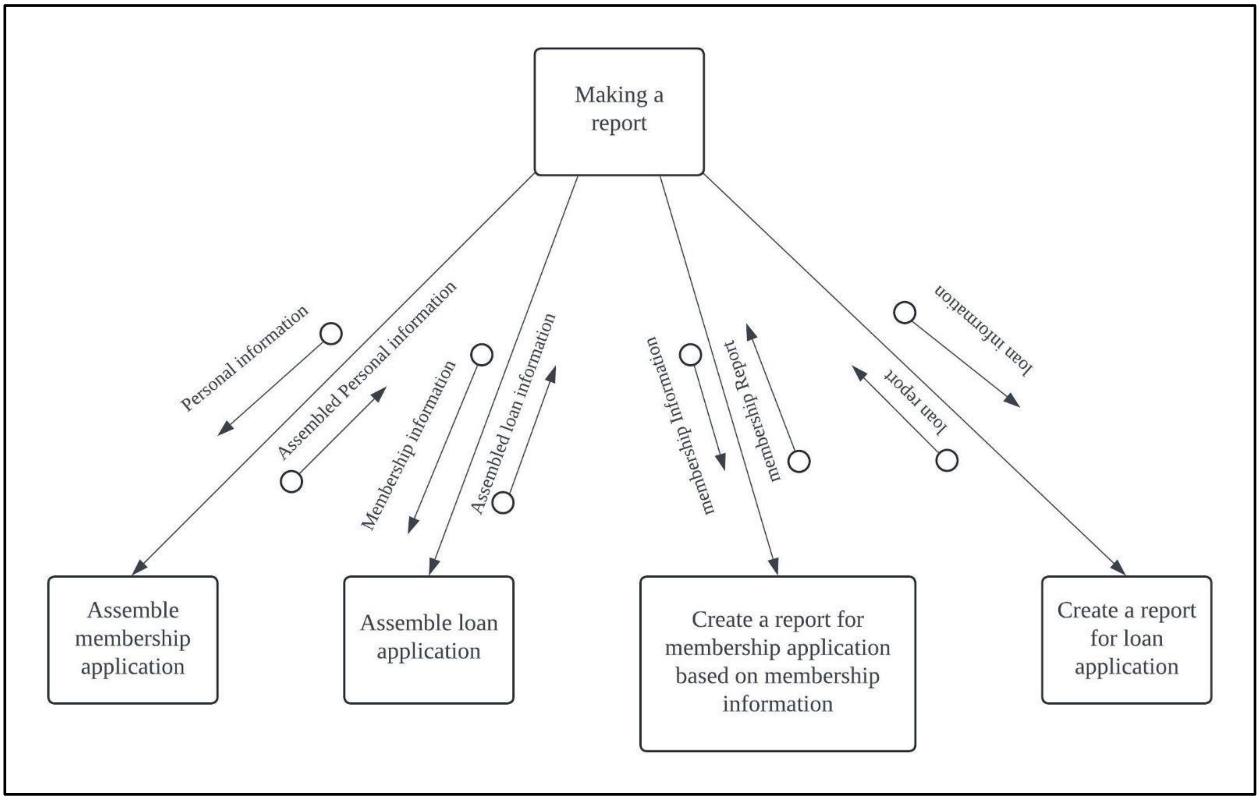


Figure 4.4.7.8

Figure 4.4.7.9 is entering and maintaining data in the system. In order to maintain the system, it has to receive the report and then plan a regular backup report for the system's future use. Then, the report information would be observed by the system security and produce the security status. In order to enter the data in the system, it has to extract the information from the report and fill it in the database.

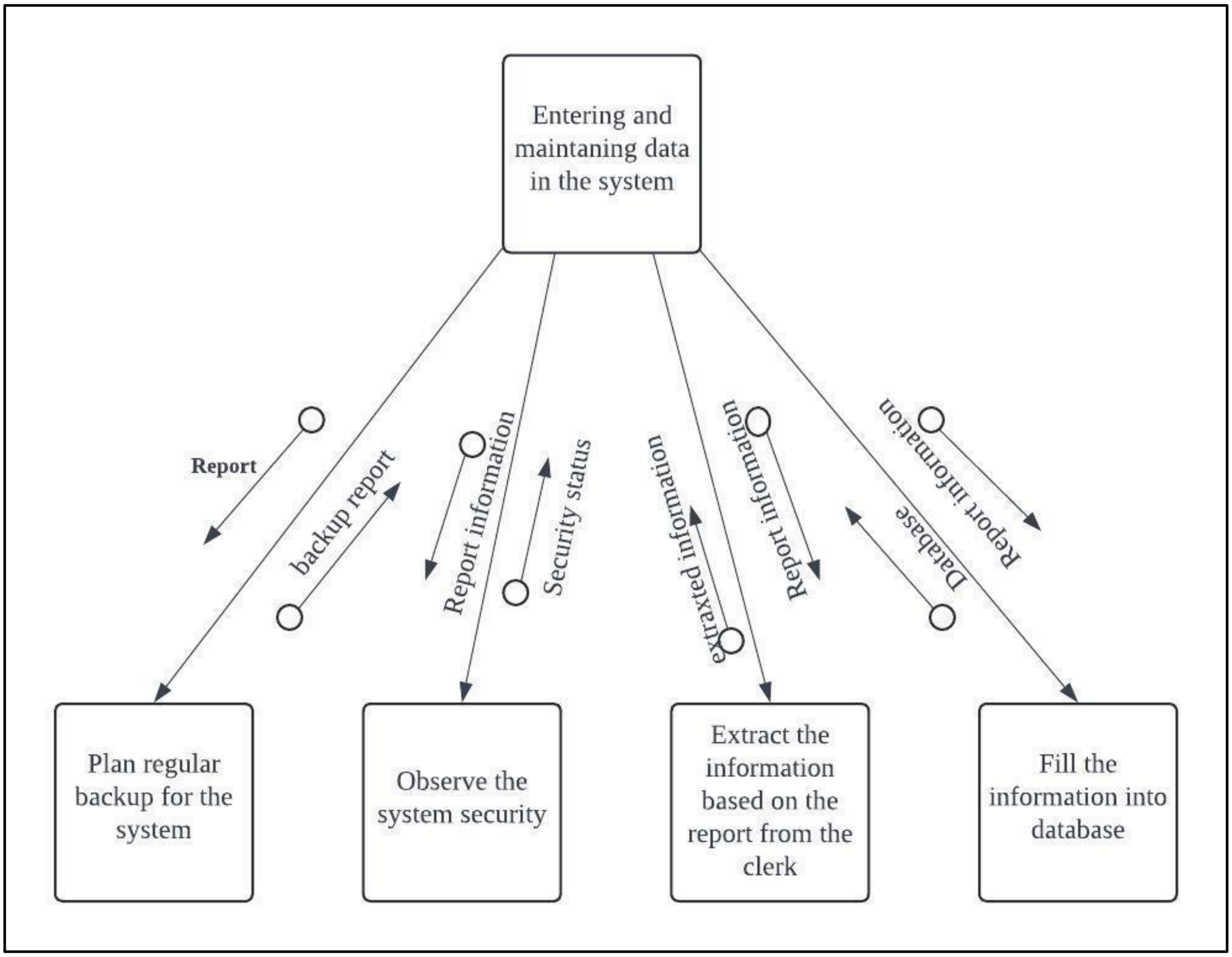


Figure 4.4.7.9

4.4.8 System Architecture

Refer to Appendix P to see the system architecture for the financial application ensures security, smooth user interaction, and efficient data management. Users access the system through web browsers or mobile apps in the Client Layer. The Security Layer protects the system with firewalls, authentication servers, intrusion detection systems, and security management tools. The Presentation Layer provides a user-friendly interface with UI elements and client-side scripts. The Business Layer handles key functions like verifying membership and loan applications, calculating repayments, and managing workflows. The Database Layer manages data storage and retrieval with a database server, data access layer, and tables for applications and repayments, along with backup and recovery systems. This structure ensures the application is secure, easy to use, and reliable.

4.4.9 Interface Design (input/Output design)

The wireframe exhibits 3 different types of user interfaces, which are applicant, clerk, and ALK. Applicants can apply for cooperative membership and also loans under the KADA cooperative in the system. Every successful application will get a digital KADA card. The clerk verifies and validates the information in the system. Clerk can access the financial statements of the applicants and the clerk also determines if a loan application is accepted or rejected based on the applicant's finances. ALK can access the automatically generated monthly report to monitor KADA business performance.

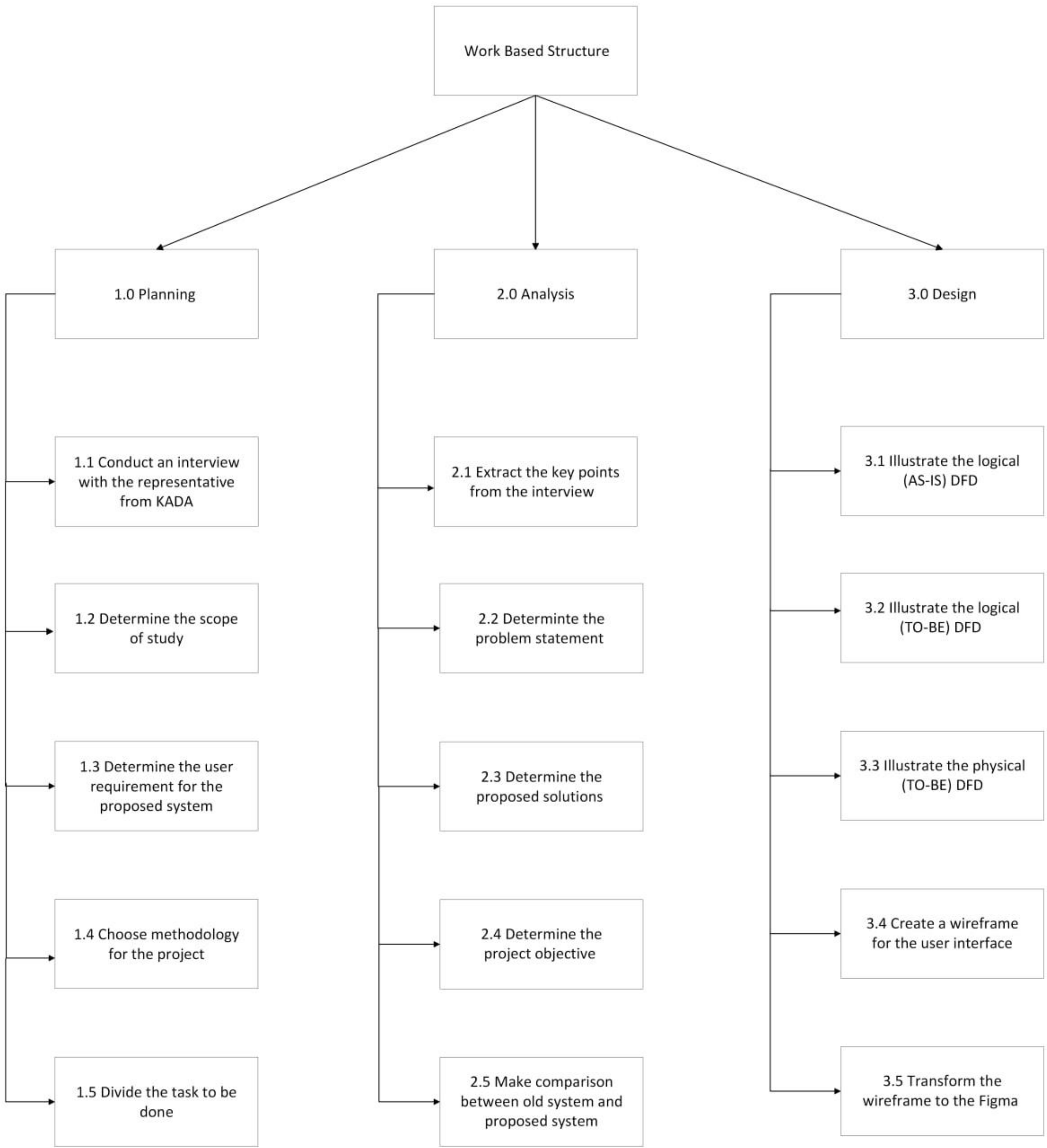
Link for figma prototype:
https://www.figma.com/design/9w5H51fZO6I0FPC68tyXJW/KADA?m=dev&node-id=77-2114&t=hIBjR0R8JWt2YzPx-1

4.5 Summary

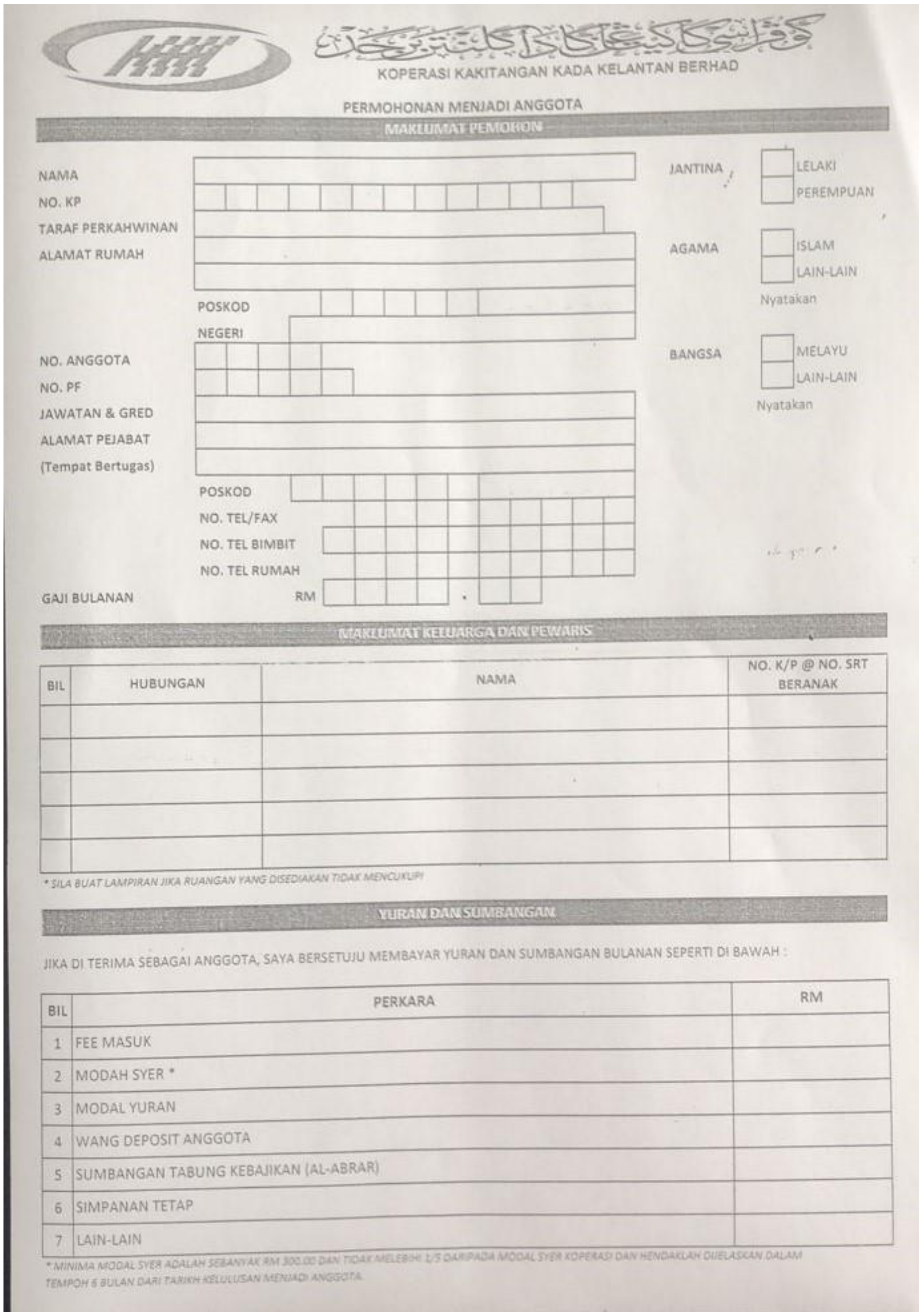
The conclusion of System Analysis and Design is about problem-solving, which calls for critical thinking, creativity, and innovation. Applying the System Development Life Cycle (SDLC) could help us a lot in developing a system, where we can look more clearly at the process and what things need to be done first so that we don't miss out on anything and the project would run smoothly. Not to forget, applying this methodology also could retain knowledge within the organization.

5.0 Appendix

Appendix A (Work Based Structure)



Appendix B Registration Form KADA



KOPERASI KAKITANGAN KADA KELANTAN BERHAD

PERMOHONAN MENJADI ANGGOTA

MAKLUMAT PEMOHON

NAMA

NO. KP

TARAF PERKAHWINAN

ALAMAT RUMAH

POSKOD

NEGERI

NO. ANGGOTA

NO. PF

JAWATAN & GRED

ALAMAT PEJABAT

(Tempat Bertugas)

POSKOD

NO. TEL/FAX

NO. TEL BIMBIT

NO. TEL RUMAH

GAJI BULANAN RM .

JANTINA: LELAKI / PEREMPUAN

AGAMA: ISLAM / LAIN-LAIN

Nyatakan

BANGSA: MELAYU / LAIN-LAIN

Nyatakan

MAKLUMAT KELUARGA DAN PEWARIS

BIL	HUBUNGAN	NAMA	NO. K/P @ NO. SRT BERANAK

* SILA BUAT LAMPIRAN JIKA RUANGAN YANG DISEDIAKAN TIDAK MENCUKUPI

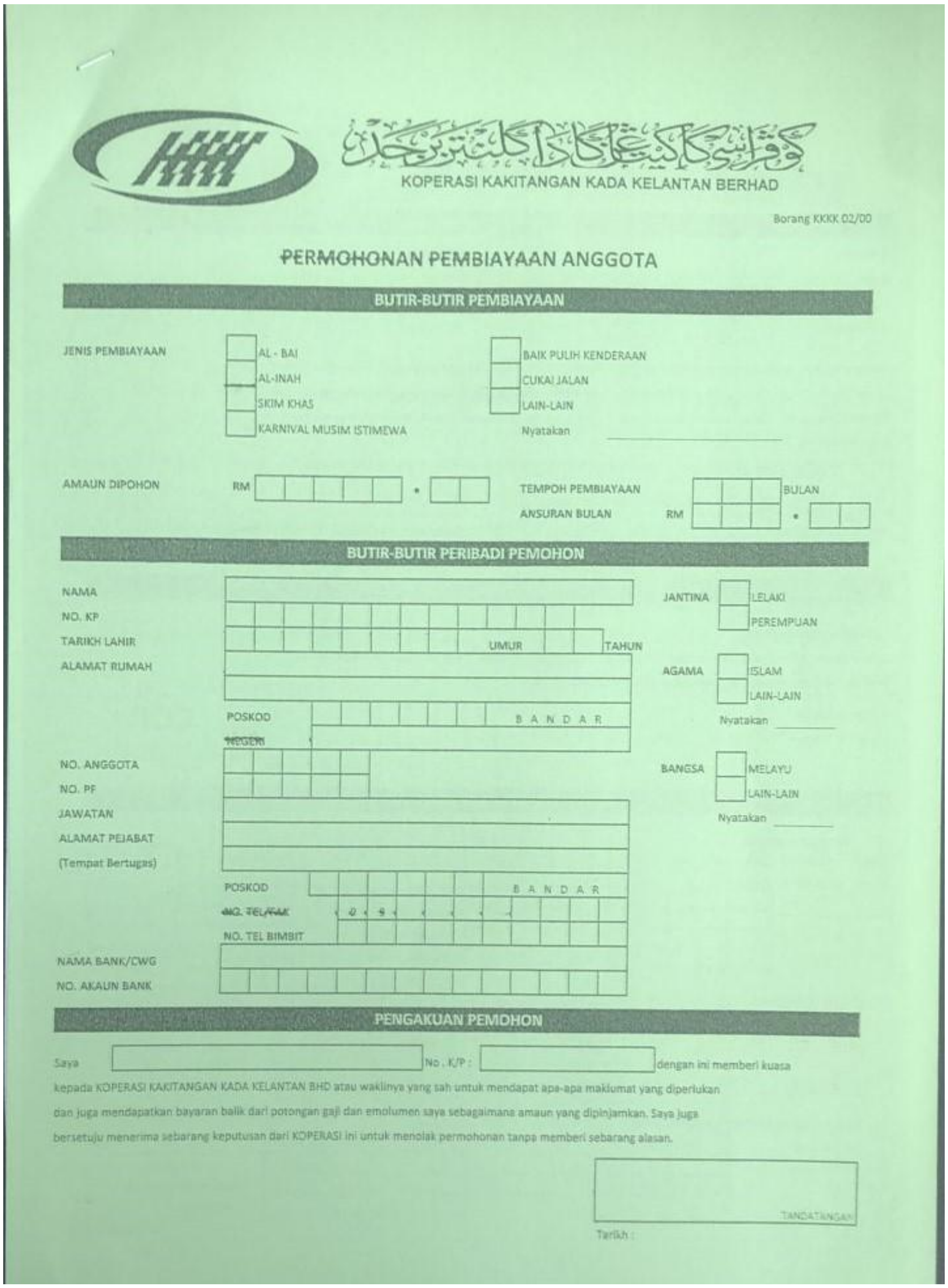
YURAN DAN SUMBANGAN

JIKA DI TERIMA SEBAGAI ANGGOTA, SAYA BERSETUJU MEMBAYAR YURAN DAN SUMBANGAN BULANAN SEPERTI DI BAWAH :

BIL	PERKARA	RM
1	FEE MASUK	
2	MODAH SYER *	
3	MODAL YURAN	
4	WANG DEPOSIT ANGGOTA	
5	SUMBANGAN TABUNG KEBAJIKAN (AL-ABRAR)	
6	SIMPANAN TETAP	
7	LAIN-LAIN	

* MINIMA MODAL SYER ADALAH SEBANYAK RM 300.00 DAN TIDAK MELEBIHI 1/5 DARIPADA MODAL SYER KOPERASI DAN HENDAKLAH DIJELASKAN DALAM TEMPOH 6 BULAN DARI TARIKH KELULUSAN MENJADI ANGGOTA

Appendix C (Loan Application Form)



KOPERASI KAKITANGAN KADA KELANTAN BERHAD

Borang KKKK 02/00

PERMOHONAN PEMBIAYAAN ANGGOTA

BUTIR-BUTIR PEMBIAYAAN

JENIS PEMBIAYAAN
- AL - BAI
- AL-INAH
- SKIM KHAS
- KARNIVAL MUSIM ISTIMEWA
- BAIK PULIH KENDERAAN
- CUKAI JALAN
- LAIN-LAIN

Nyatakan

AMAUN DIPOHON RM .

TEMPOH PEMBIAYAAN BULAN

ANSURAN BULAN RM .

BUTIR-BUTIR PERIBADI PEMOHON

NAMA

NO. KP

TARIKH LAHIR UMUR TAHUN

ALAMAT RUMAH

POSKOD BANDAR

~~NEGERI~~

JANTINA
- LELAKI
- PEREMPUAN

AGAMA
- ISLAM
- LAIN-LAIN

Nyatakan

NO. ANGGOTA

NO. PF

JAWATAN

ALAMAT PEJABAT (Tempat Bertugas)

POSKOD BANDAR

~~NO. TEL/FAX~~ 0 9

NO. TEL BIMBIT

BANGSA
- MELAYU
- LAIN-LAIN

Nyatakan

NAMA BANK/CWG

NO. AKAUN BANK

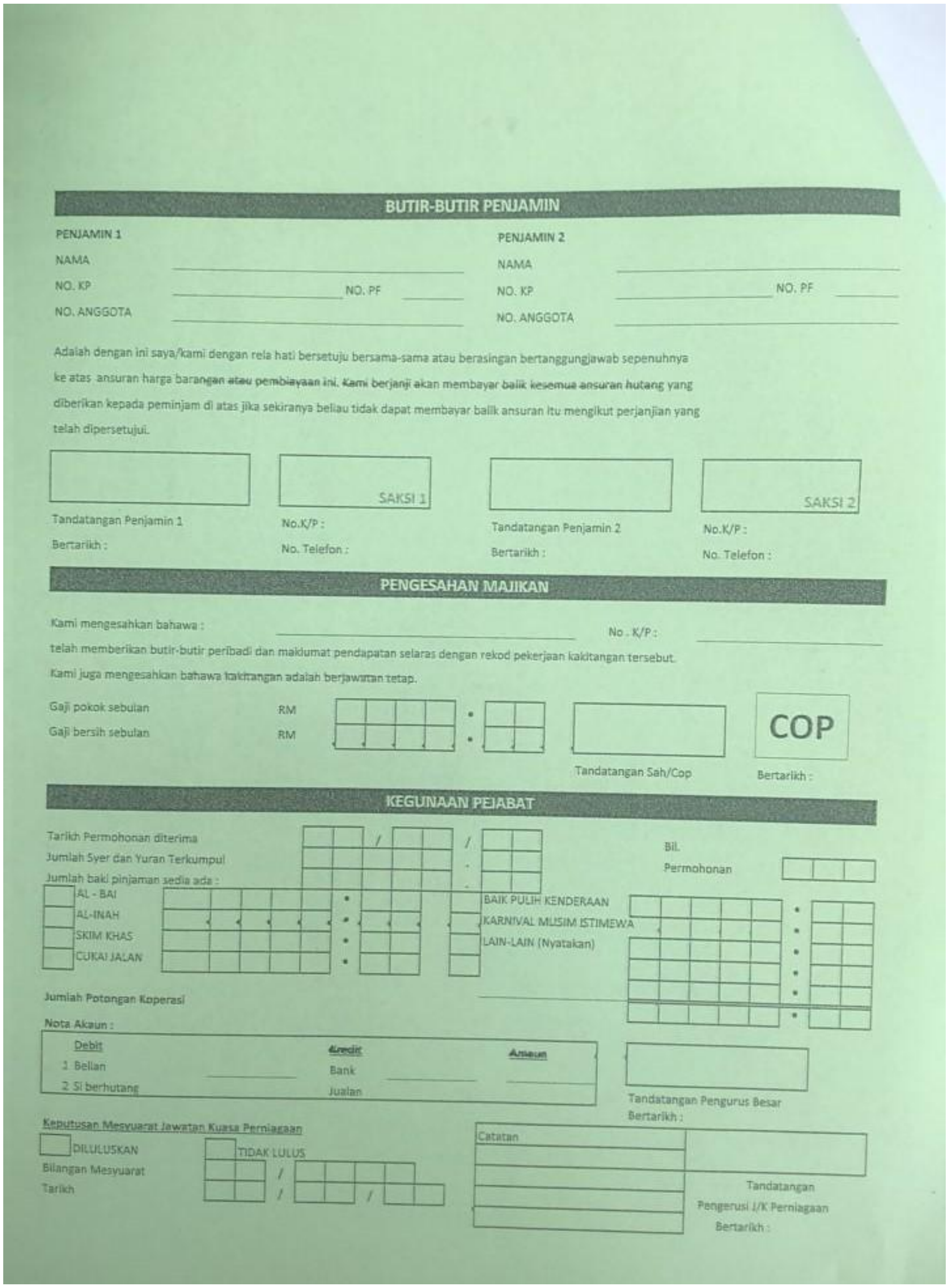
PENGAKUAN PEMOHON

Saya No . K/P : dengan ini memberi kuasa kepada KOPERASI KAKITANGAN KADA KELANTAN BHD atau wakilnya yang sah untuk mendapat apa-apa maklumat yang diperlukan dan juga mendapatkan bayaran balik dari potongan gaji dan emolumen saya sebagaimana amaun yang dipinjamkan. Saya juga bersetuju menerima sebarang keputusan dari KOPERASI ini untuk menolak permohonan tanpa memberi sebarang alasan.

TANDATANGAN

Tarikh :

Appendix D (Loan Application Form)



BUTIR-BUTIR PENJAMIN

PENJAMIN 1

NAMA

NO. KP NO. PF

NO. ANGGOTA

PENJAMIN 2

NAMA

NO. KP NO. PF

NO. ANGGOTA

Adalah dengan ini saya/kami dengan rela hati bersetuju bersama-sama atau berasingan bertanggungjawab sepenuhnya ke atas ansuran harga barangan atau pembiayaan ini. Kami berjanji akan membayar balik kesemua ansuran hutang yang diberikan kepada peminjam di atas jika sekiranya beliau tidak dapat membayar balik ansuran itu mengikut perjanjian yang telah dipersetujui.

Tandatangan Penjamin 1

Bertarikh :

SAKSI 1

No.K/P :

No. Telefon :

Tandatangan Penjamin 2

Bertarikh :

SAKSI 2

No.K/P :

No. Telefon :

PENGESAHAN MAJIKAN

Kami mengesahkan bahawa : No . K/P :

telah memberikan butir-butir peribadi dan maklumat pendapatan selaras dengan rekod pekerjaan kakitangan tersebut.

Kami juga mengesahkan bahawa kakitangan adalah berjawatan tetap.

Gaji pokok sebulan RM .

Gaji bersih sebulan RM .

Tandatangan Sah/Cop

COP

Bertarikh :

KEGUNAAN PEJABAT

Tarikh Permohonan diterima / /

Jumlah Syer dan Yuran Terkumpul

Jumlah baki pinjaman sedia ada :

AL - BAI

AL-INAH

SKIM KHAS

CUKAI JALAN

BAIK PULIH KENDERAAN

KARNIVAL MUSIM ISTIMEWA

LAIN-LAIN (Nyatakan)

Bil.

Permohonan

Jumlah Potongan Koperasi

Nota Akaun :

Debit	Kredit	Amaun
1 Belian	Bank	
2 Si berhutang	Jualan	

Tandatangan Pengurus Besar

Bertarikh :

Keputusan Mesyuarat Jawatan Kuasa Perniagaan

DILULUSKAN

TIDAK LULUS

Bilangan Mesyuarat /

Tarikh / /

Catatan

Tandatangan

Pengerusi J/K Perniagaan

Bertarikh :

Appendix E (Monthly Cooperative Member Financial Report)

KOPERASI KAKITANGAN KADA KELANTAN BERHAD

NAMA: MOHD ROSLAN B ISMAIL

NO. K/P: 740807-03-5479 **NO. PF:** 2273

NO. AHLI: 1579

Tuan/Puan,

PENGESAHAN PENYATA KEWANGAN AHLI KOPERASI KAKITANGAN KADA KELANTAN BERHAD BAGI TAHUN BERAKHIR 30 JUN 2023.

Untuk penentuan Juruaudit, kami dengan ini menyatakan bagi akaun tuan/puan adalah sebagaimana berikut:

MAKLUMAT SAHAM AHLI:

Modah Syer:	Modal Yuran:	Simpanan Tetap:	Tabung Anggota:	Simpanan Anggota:
RM[illegible]	RM[illegible]	RM[illegible]	RM[illegible]	RM[illegible]

MAKLUMAT PINJAMAN AHLI:

Al-Bai:	Al-Innah:	B/Pulih Kenderaan:	Road Tax& Insuran:	Khas:	Al-Qadrul Hassan:
RM0.00	RM0.00	RM0.00	RM0.00	RM0.00	RM0.00

PENGESAHAN BAGI PENYATA KEWANGAN
AHLI KOPERASI KAKITANGAN KADA KELANTAN BERHAD BAGI TAHUN BERAKHIR 30 JUN 2023

Saya MOHD ROSLAN B ISMAIL No. Ahli: 1579 mengesahkan bahawa Penyata Kewangan Koperasi Kakitangan KADA Kelantan Berhad bagi tahun berakhir 30 Jun 2023 adalah benar:

Tandatangan: ....
Tarikh: ....

☐ Setuju

☐ Tidak Setuju...
(Nyatakan sebabnya)

Nota:
1. Sila tandakan (/) di salah satu kotak diatas.
2. Sekiranya pihak Koperasi tidak menerima sebarang maklumbalas daripada tuan/puan sehingga 30 Jun 2023. maka pengesahan penyata ini dianggap betul dan tuan/puan bersetuju.

Appendix F (Loan Repayment Schedule)

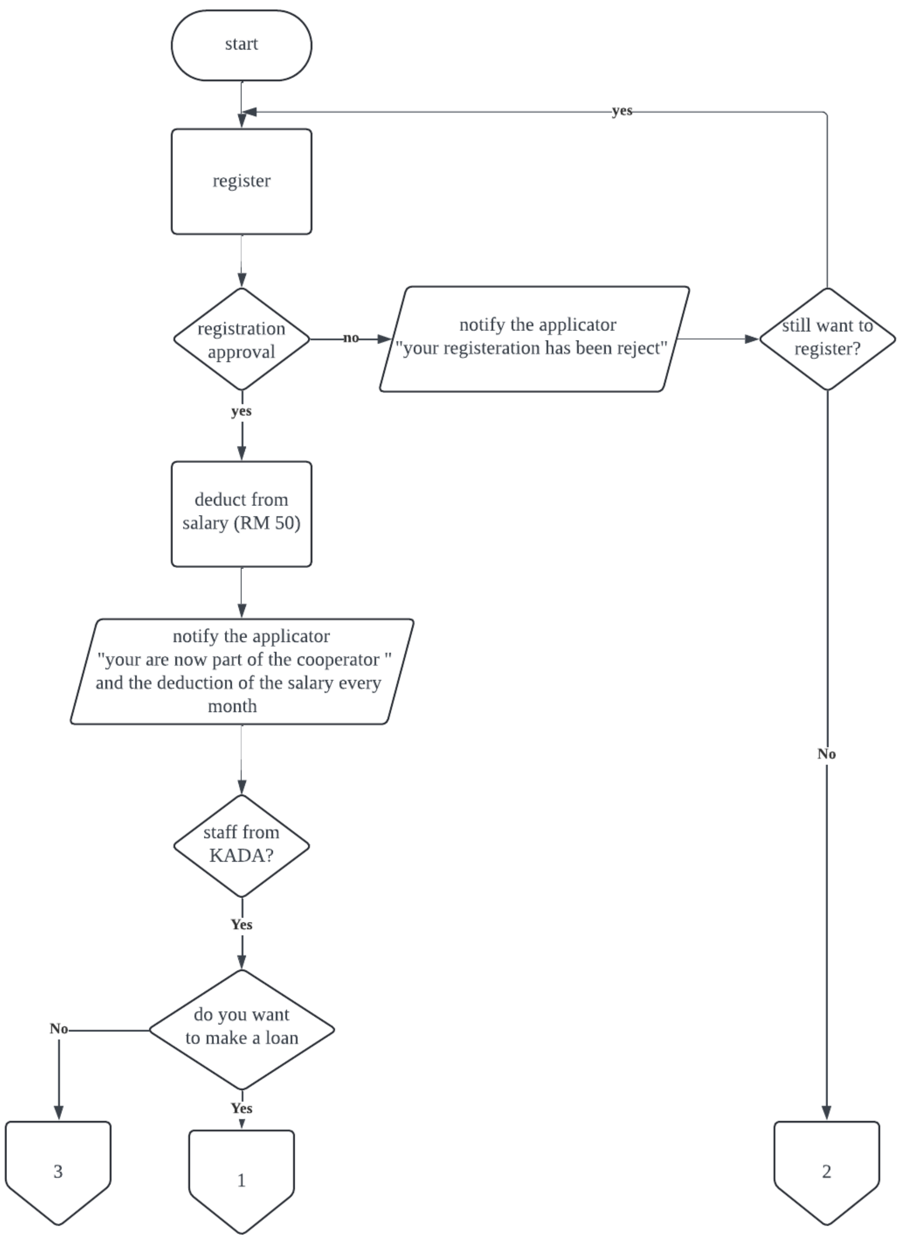
JADUAL PEMBAYARAN BALIK PEMBIAYAAN SKIM AL-BAI'UBITHAMAN AAJIL/BAI AL-INAH

KADAR KEUNTUNGAN	4.20%					
TEMPOH (TAHUN)	1	2	3	4	5	6
TEMPOH (BULANAN)	12	24	36	48	60	72
JUMLAH PEMBIAYAAN	ANSURAN BULANAN					
1,000.00	86.83	45.17	31.28	24.33	20.17	17.39
2,000.00	173.67	90.33	62.56	48.67	40.33	34.78
3,000.00	260.50	135.50	93.83	73.00	60.50	52.17
4,000.00	347.33	180.67	125.11	97.33	80.67	69.56
5,000.00	434.17	225.83	156.39	121.67	100.83	86.94
6,000.00	521.00	271.00	187.67	146.00	121.00	104.33
7,000.00	607.83	316.17	218.94	170.33	141.17	121.72
8,000.00	694.67	361.33	250.22	194.67	161.33	139.11
9,000.00	781.50	406.50	281.50	219.00	181.50	156.50
10,000.00	868.33	451.67	312.78	243.33	201.67	173.89
11,000.00	955.17	496.83	344.06	267.67	221.83	191.28
12,000.00	1042.00	542.00	375.33	292.00	242.00	208.67
13,000.00	1128.83	587.17	406.61	316.33	262.17	226.06
14,000.00	1215.67	632.33	437.89	340.67	282.33	243.44
15,000.00	1302.50	677.50	469.17	365.00	302.50	260.83
16,000.00	1389.33	722.67	500.44	389.33	322.67	278.22
17,000.00	1476.17	767.83	531.72	413.67	342.83	295.61
18,000.00	1563.00	813.00	563.00	438.00	363.00	313.00
19,000.00	1649.83	858.17	594.28	462.33	383.17	275.06
20,000.00	1736.67	903.33	625.56	486.67	403.33	347.78

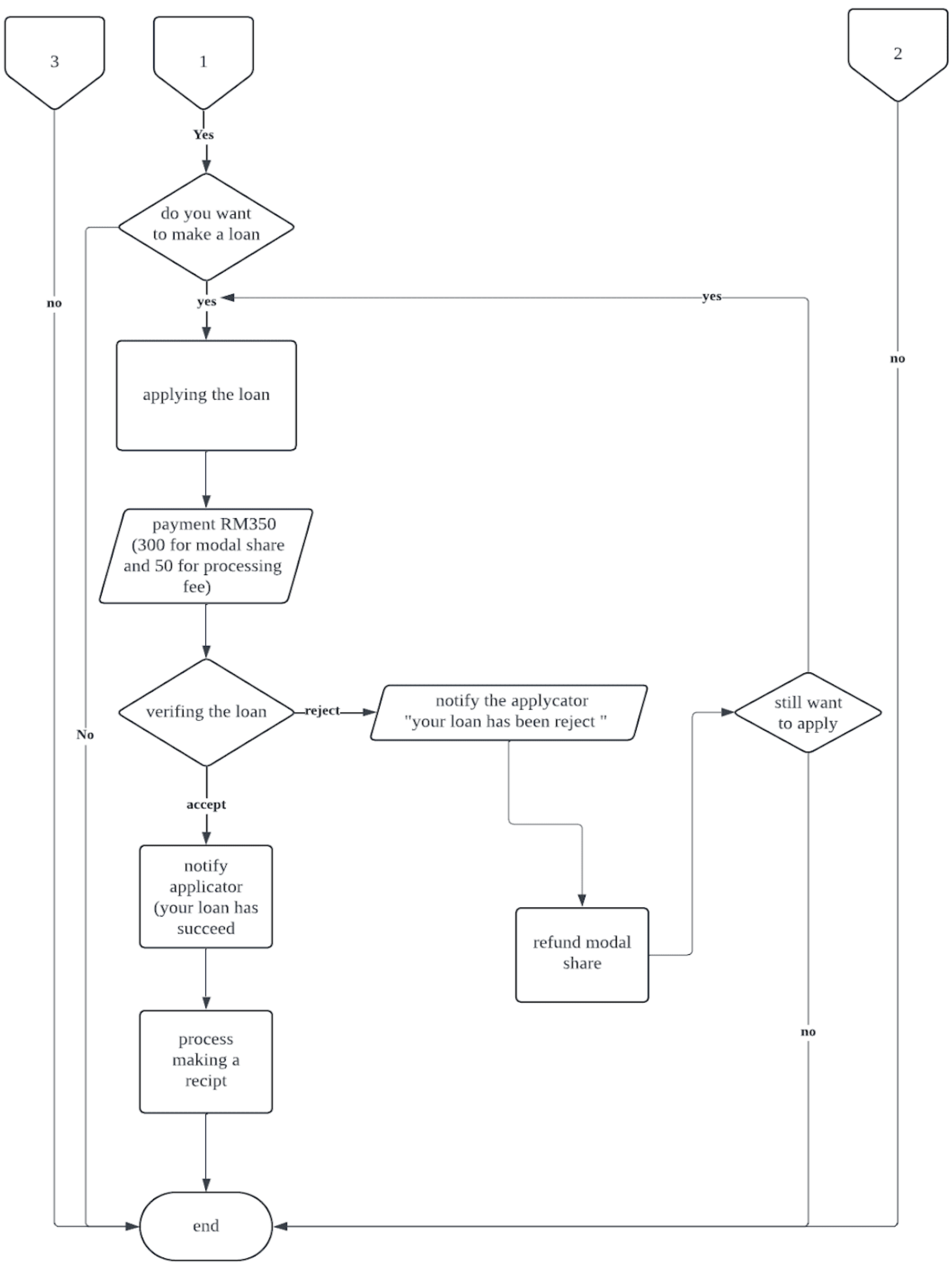
Dikemaskini 10 Mei 2023

Mesyuarat Lembaga bil 3/2023 7 Mei 2023

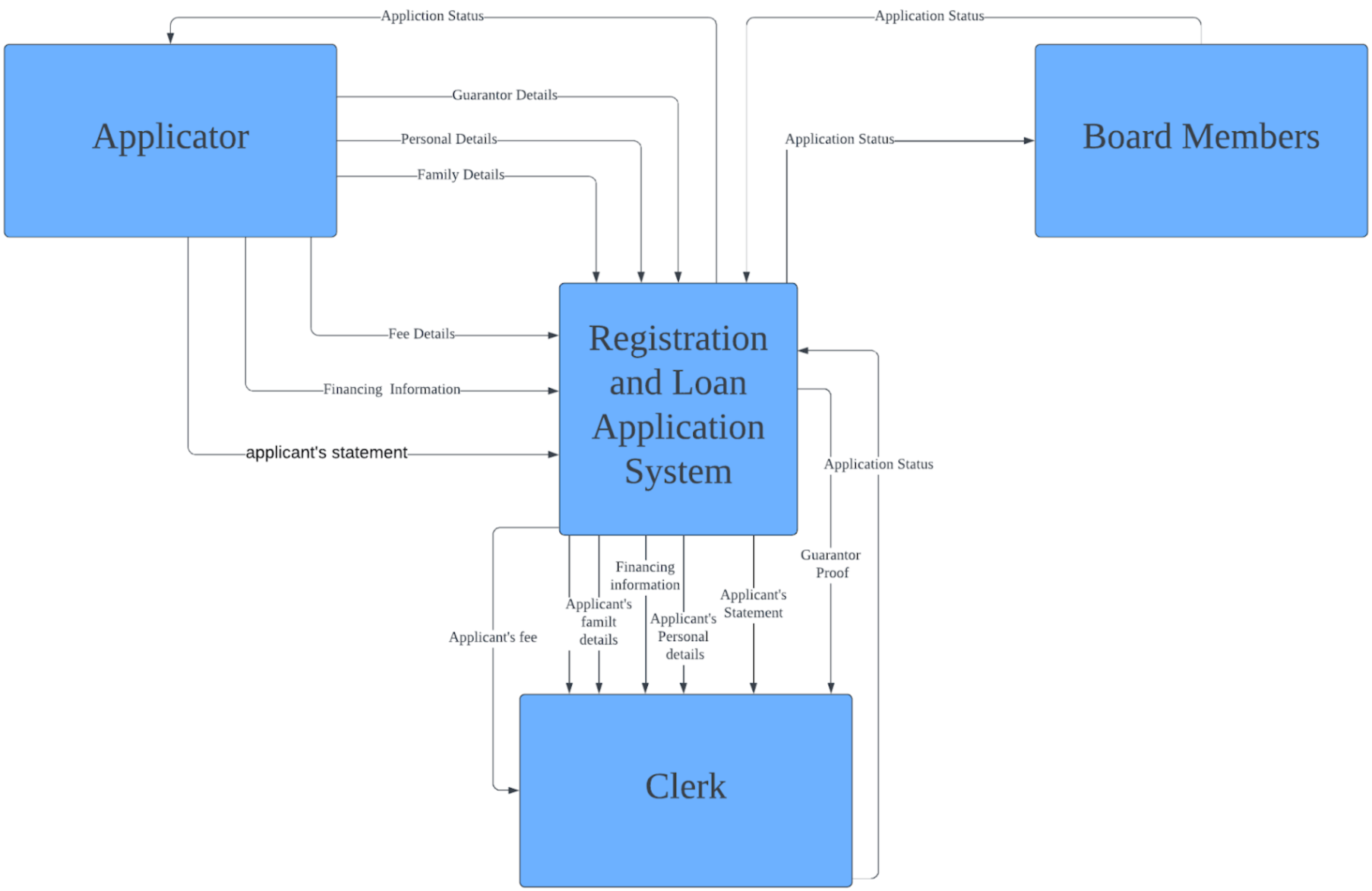
Appendix G (flowchart of the workflow)



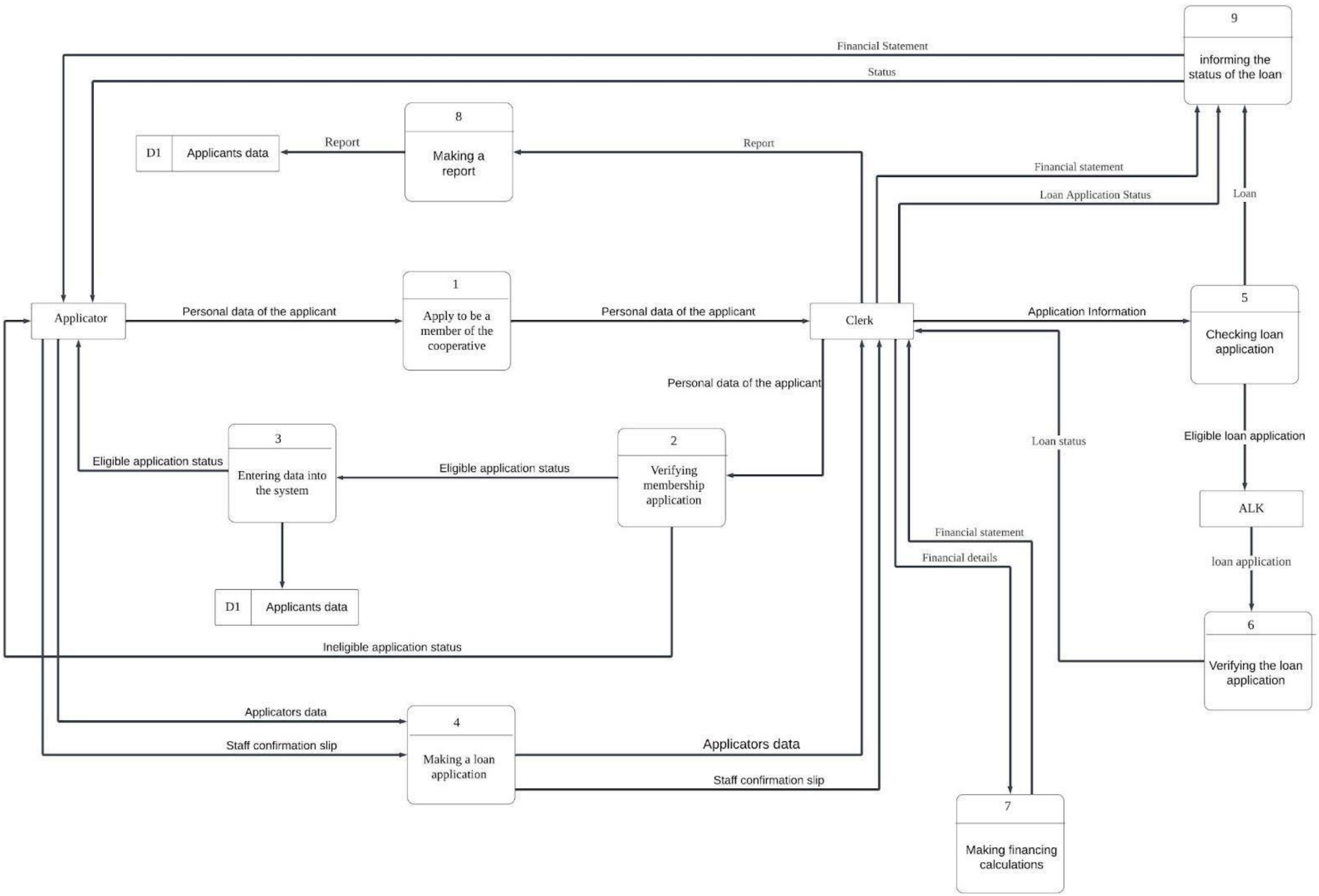
Appendix H (Flowchart of the workflow)



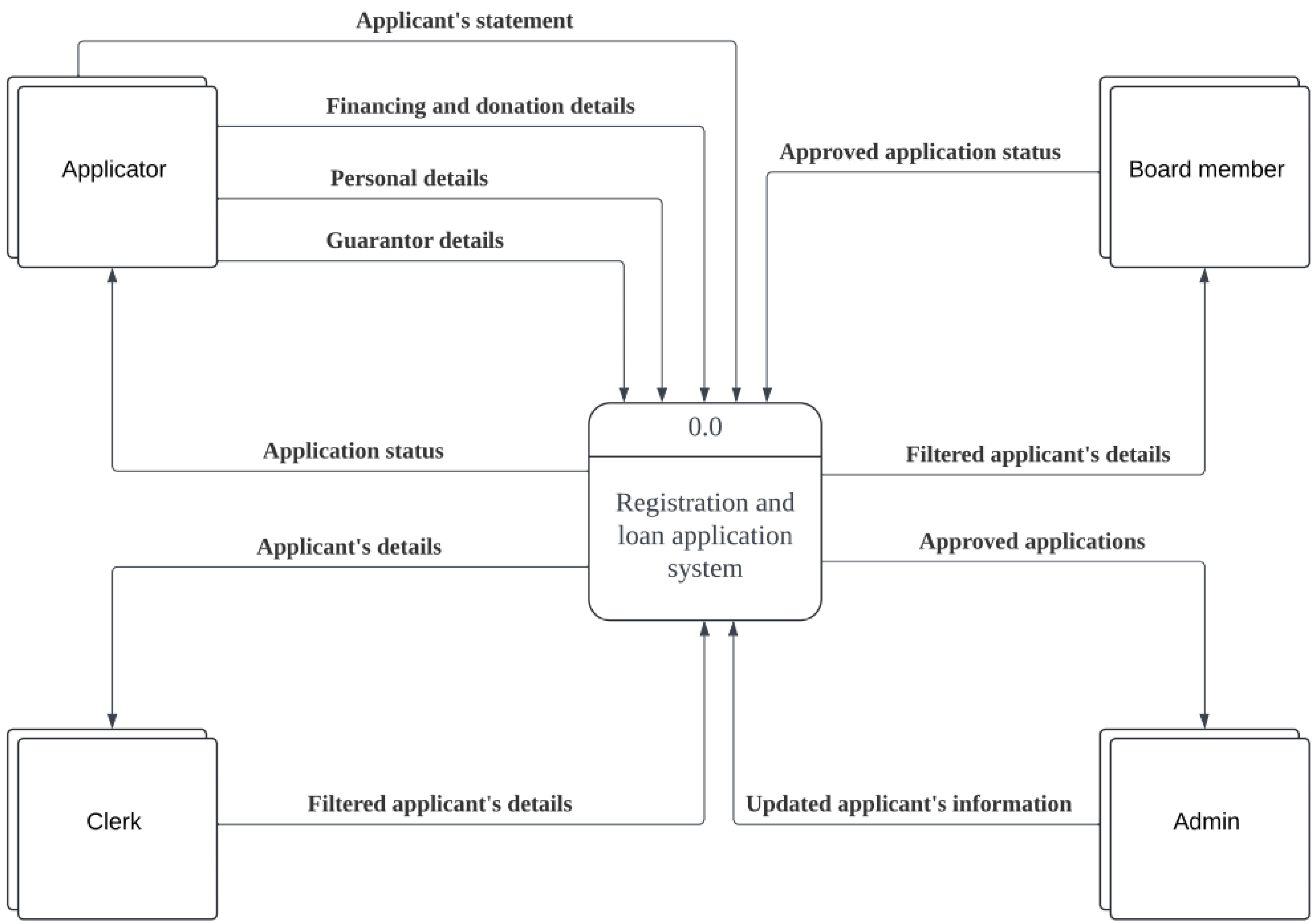
Appendix I (context diagram of the system(AS-IS))



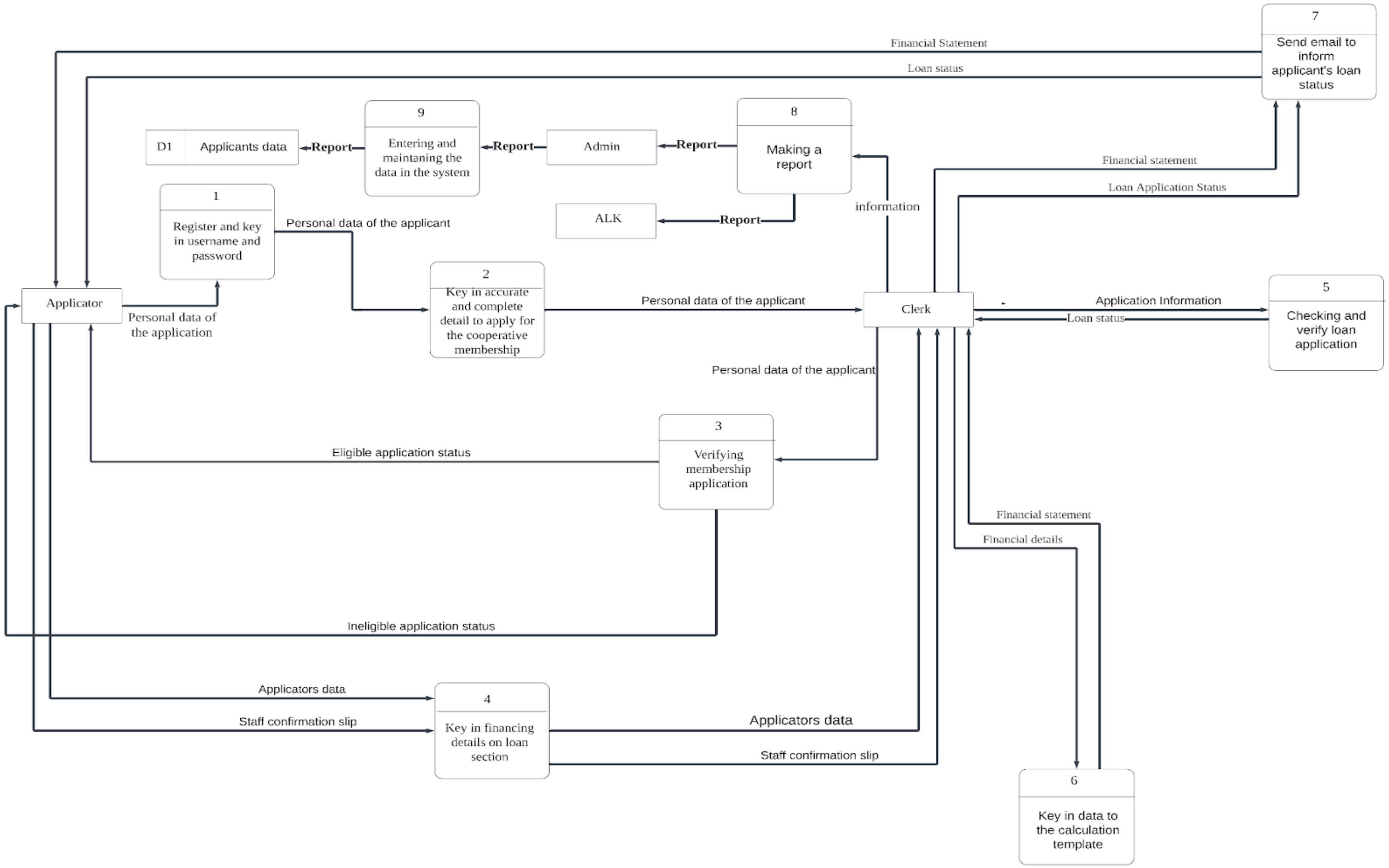
Appendix J(Diagram zero of the system (AS-IS))



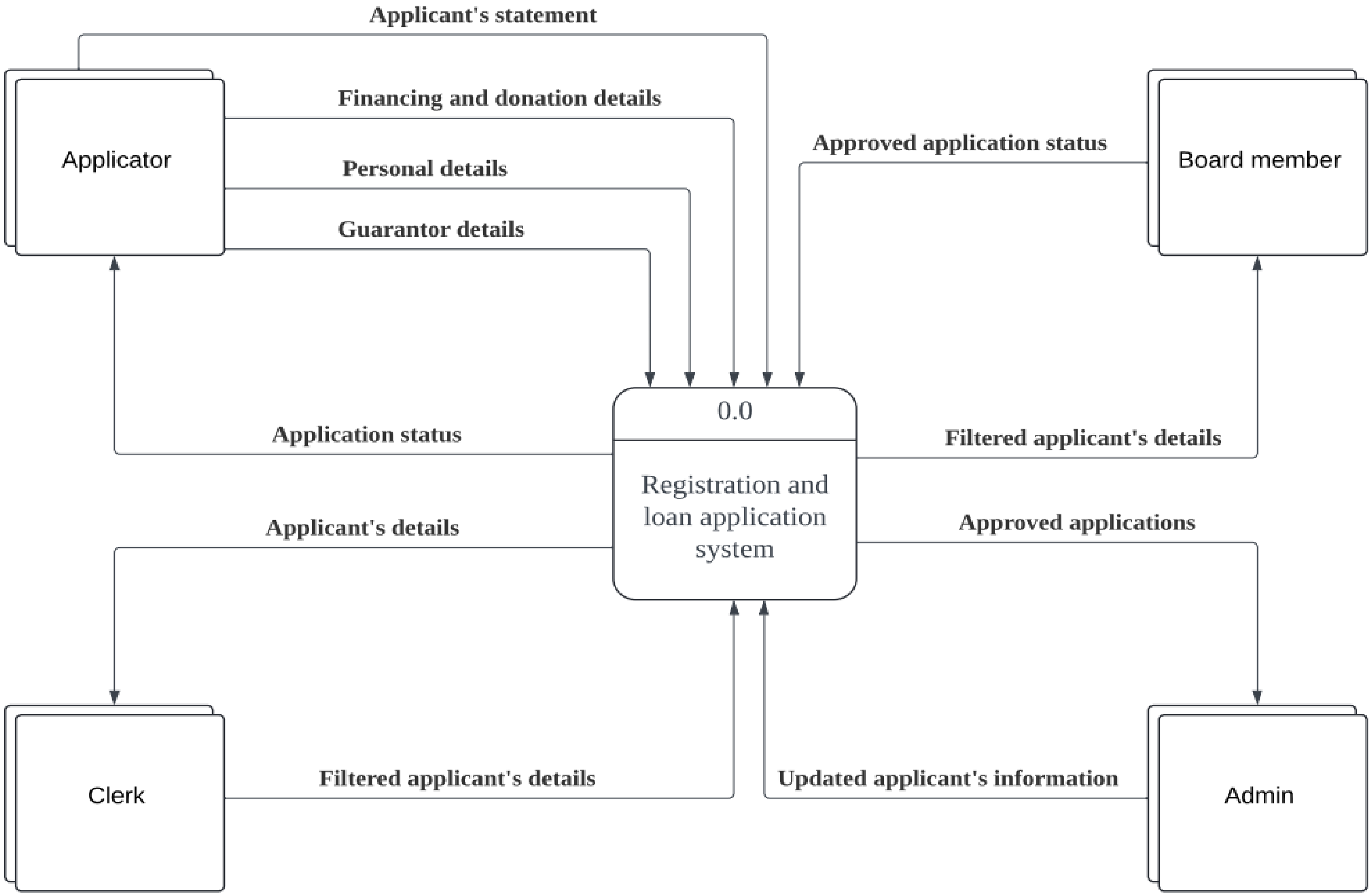
Appendix K (context diagram TO-B)



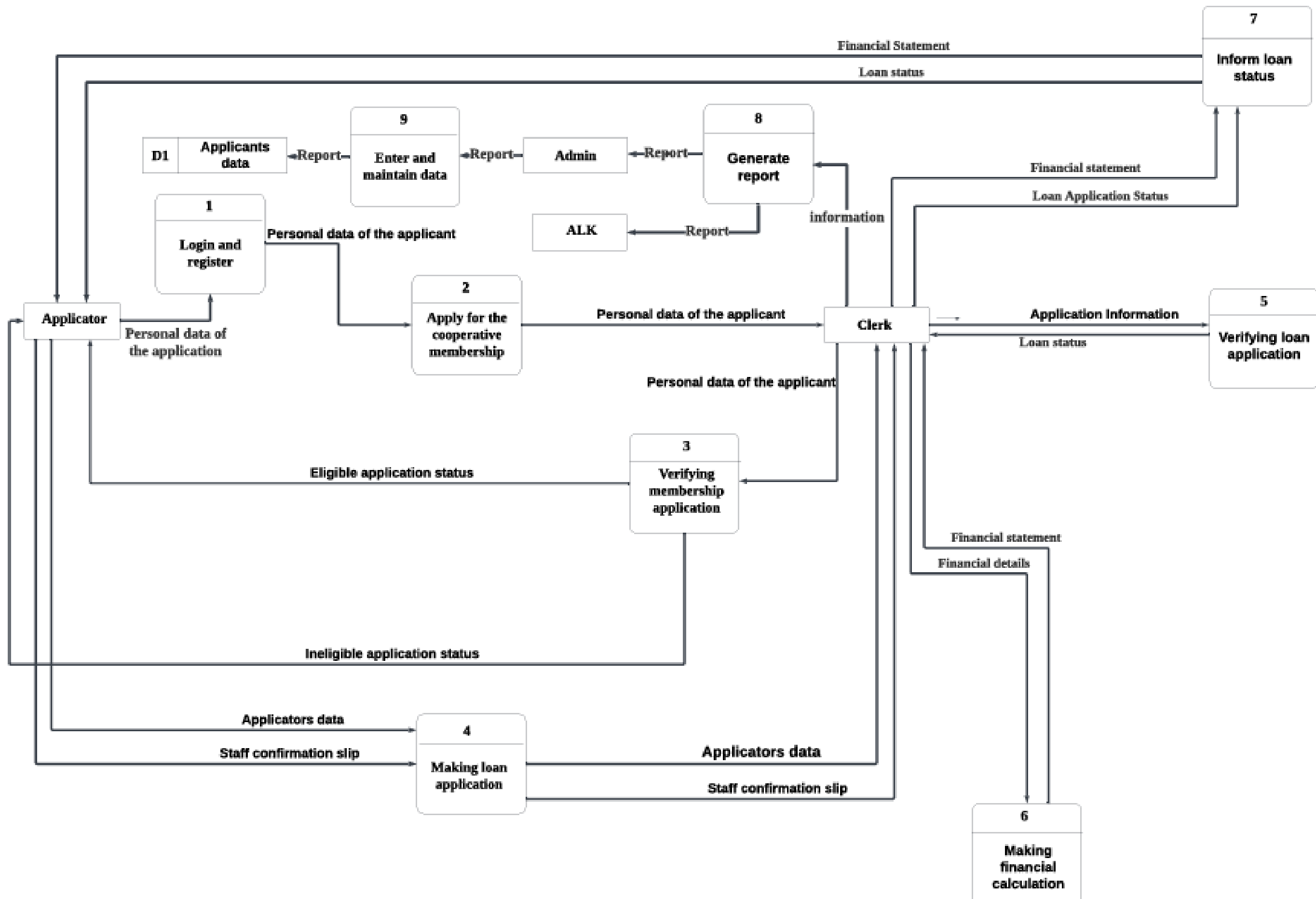
Appendix L Physical Diagram zero TO-BE



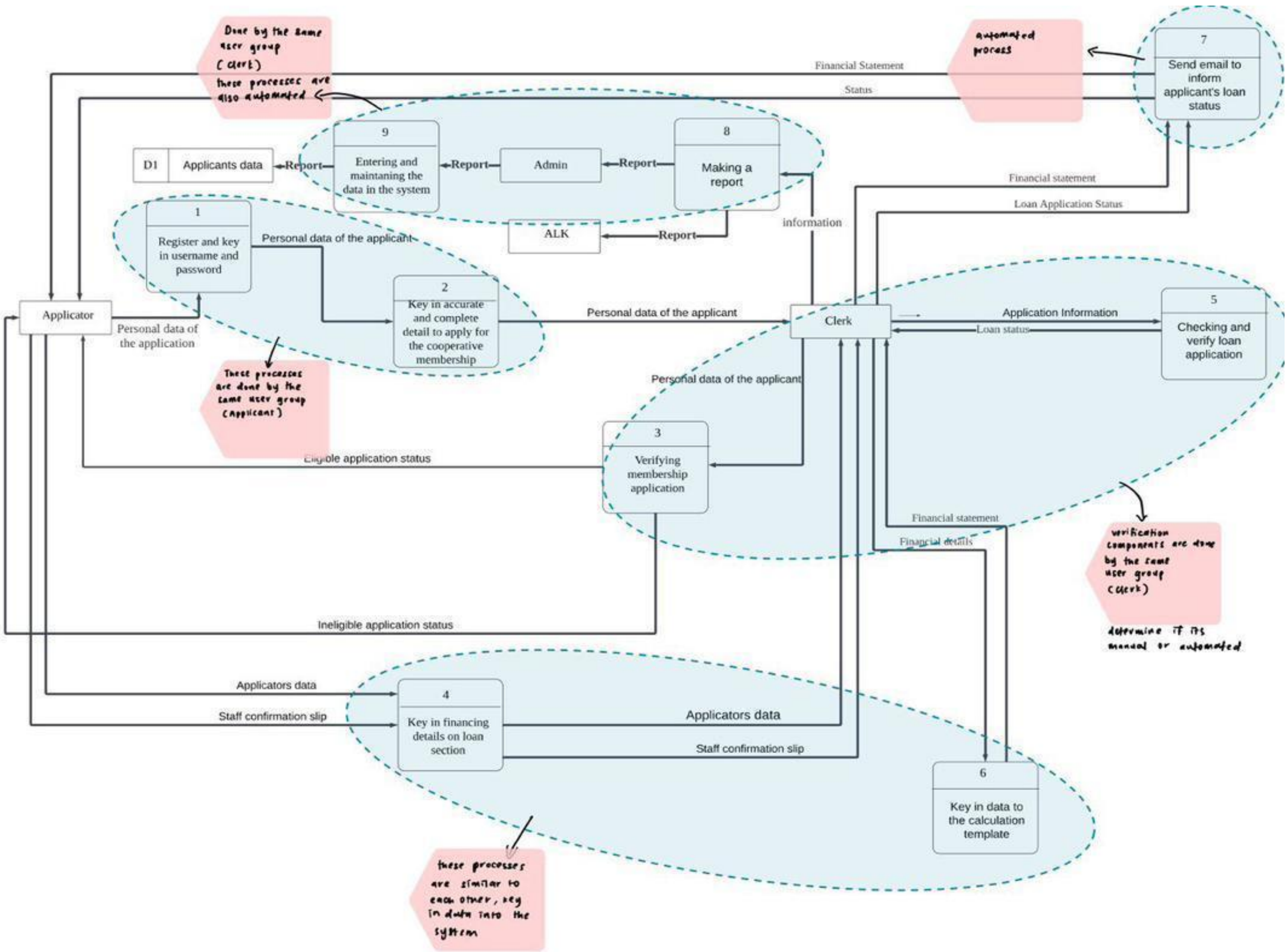
Appendix M (Context diagram physical DFD (TO-BE) system)



Appendix N (Level zero physical DFD (TO-BE) system)



Appendix O (Partitioning diagram)



Appendix P (System Architechture)